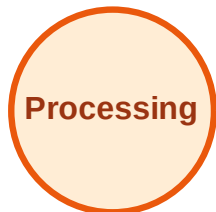


BFS Traversal

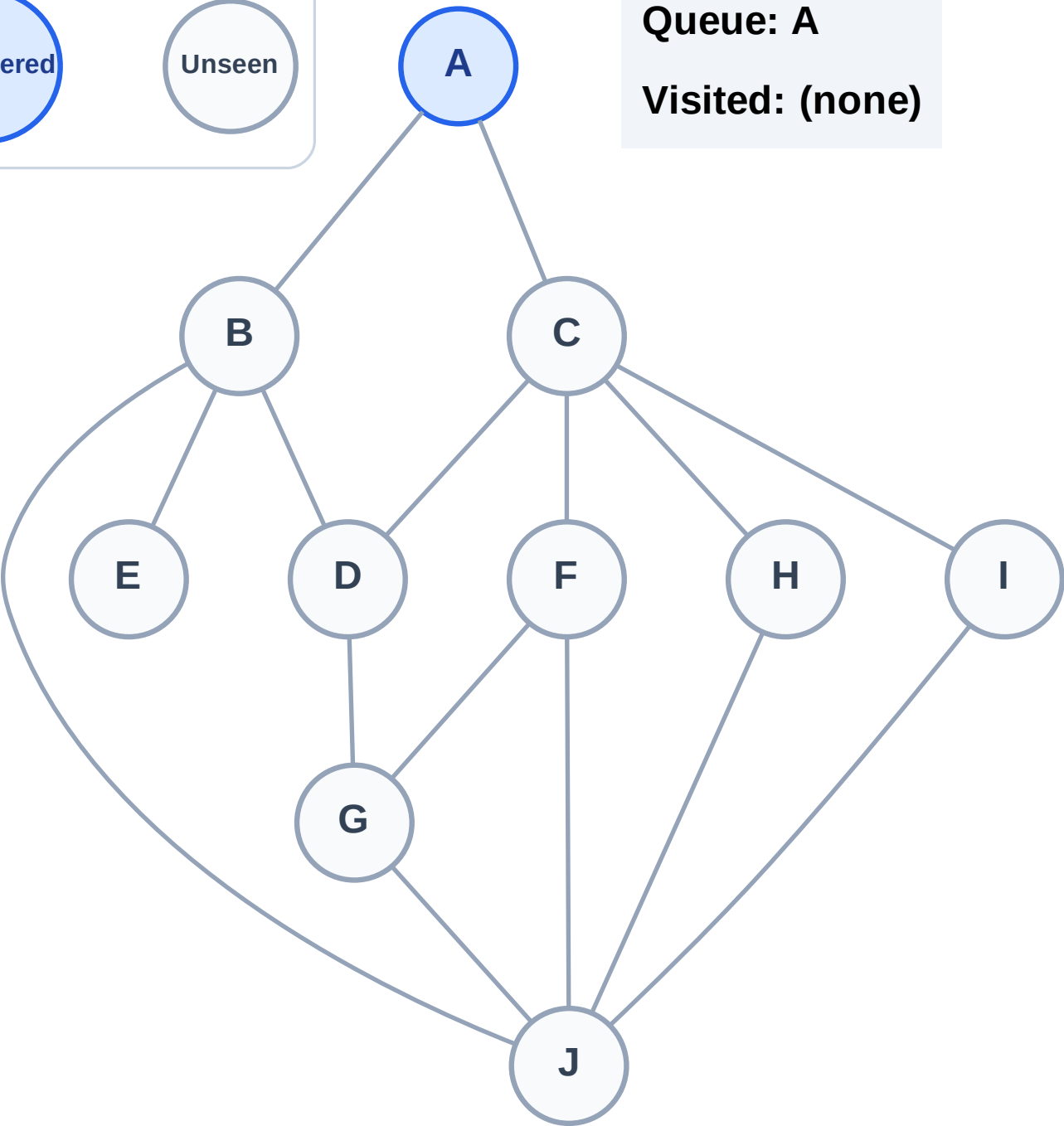
Step 1: Start at A

Legend



Queue: A

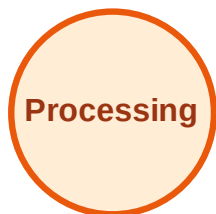
Visited: (none)



BFS Traversal

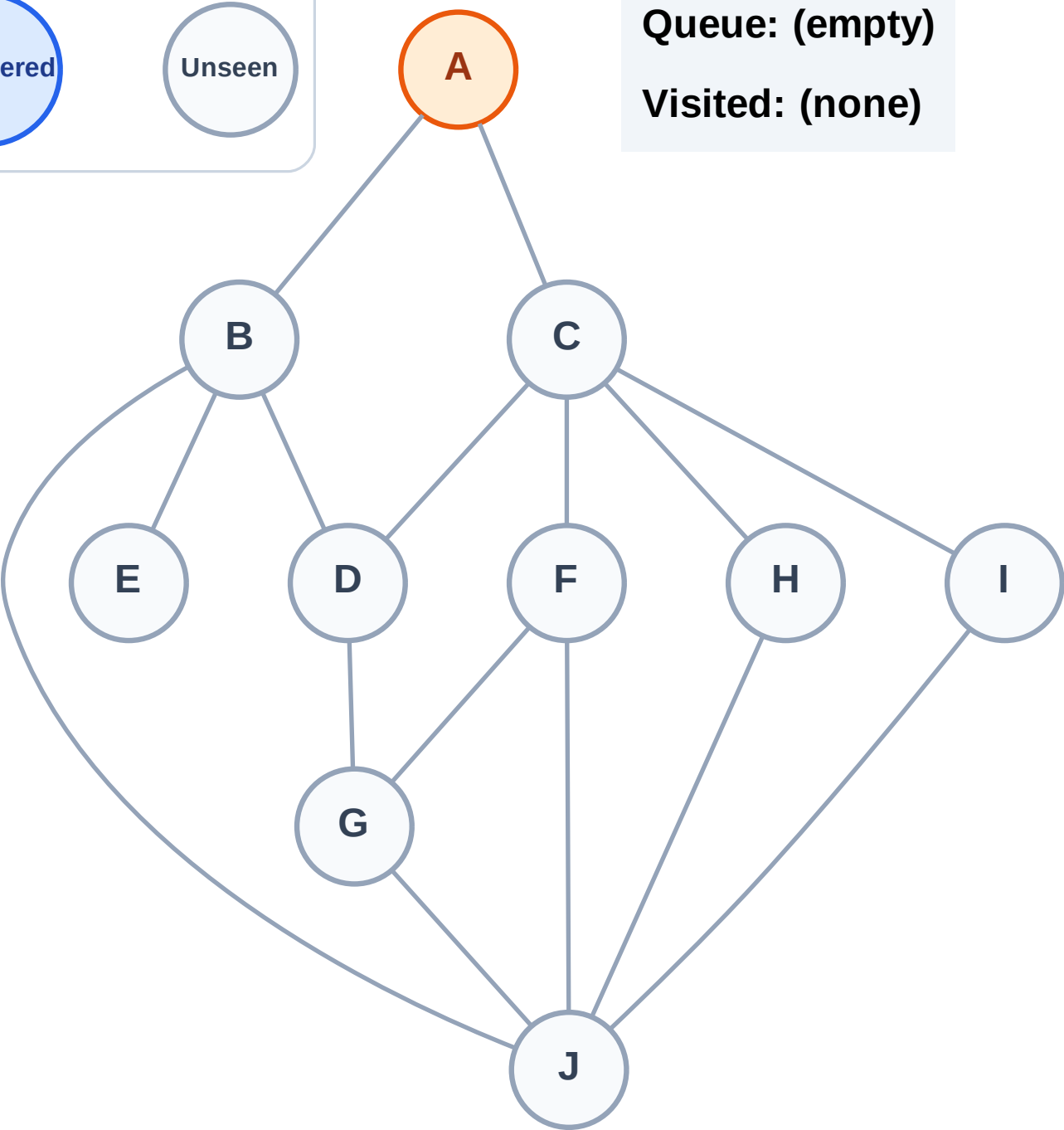
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

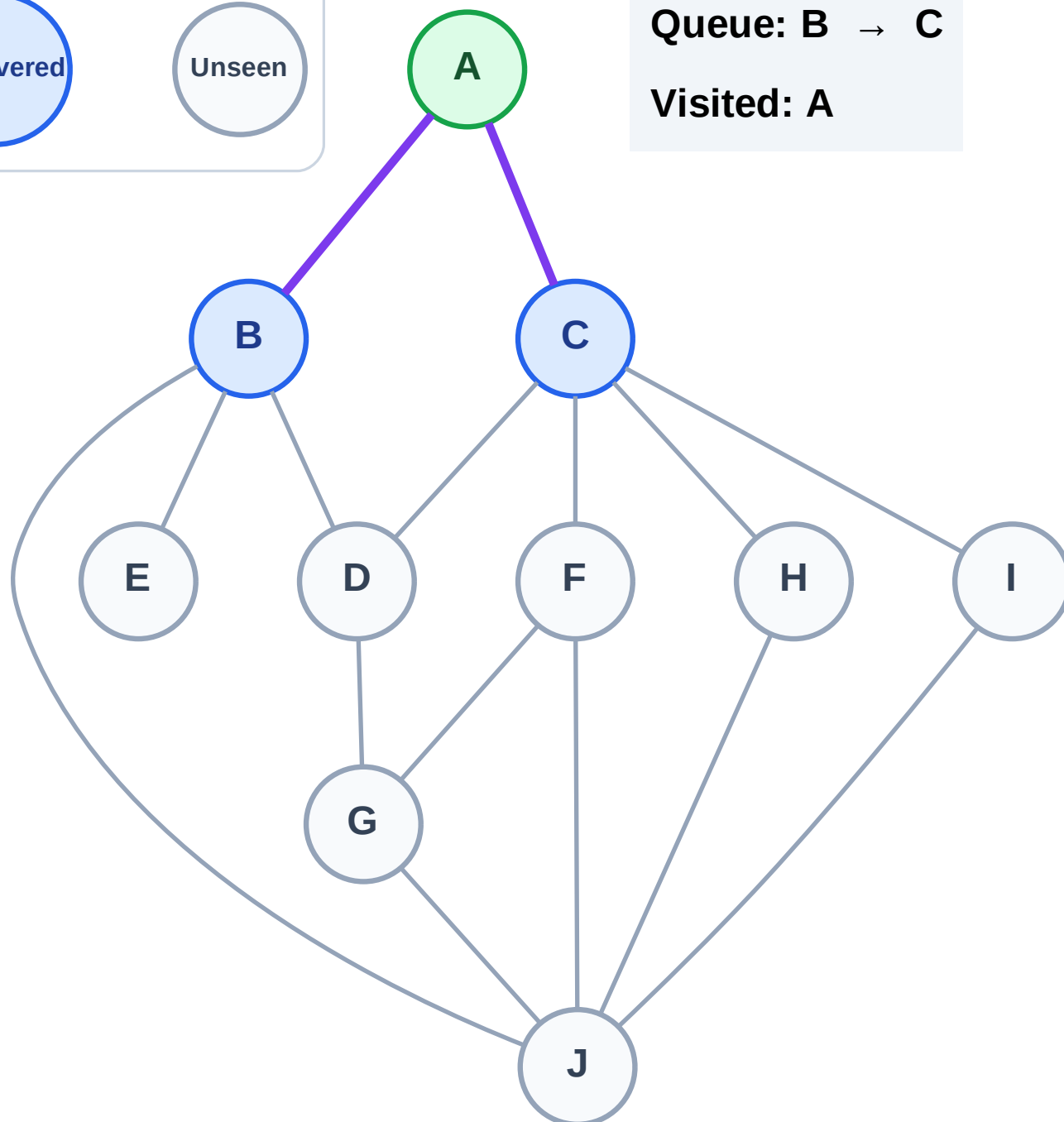
Step 3: Discover B, C from A

Legend



Queue: B → C

Visited: A



BFS Traversal

Step 4: Process node B

Legend

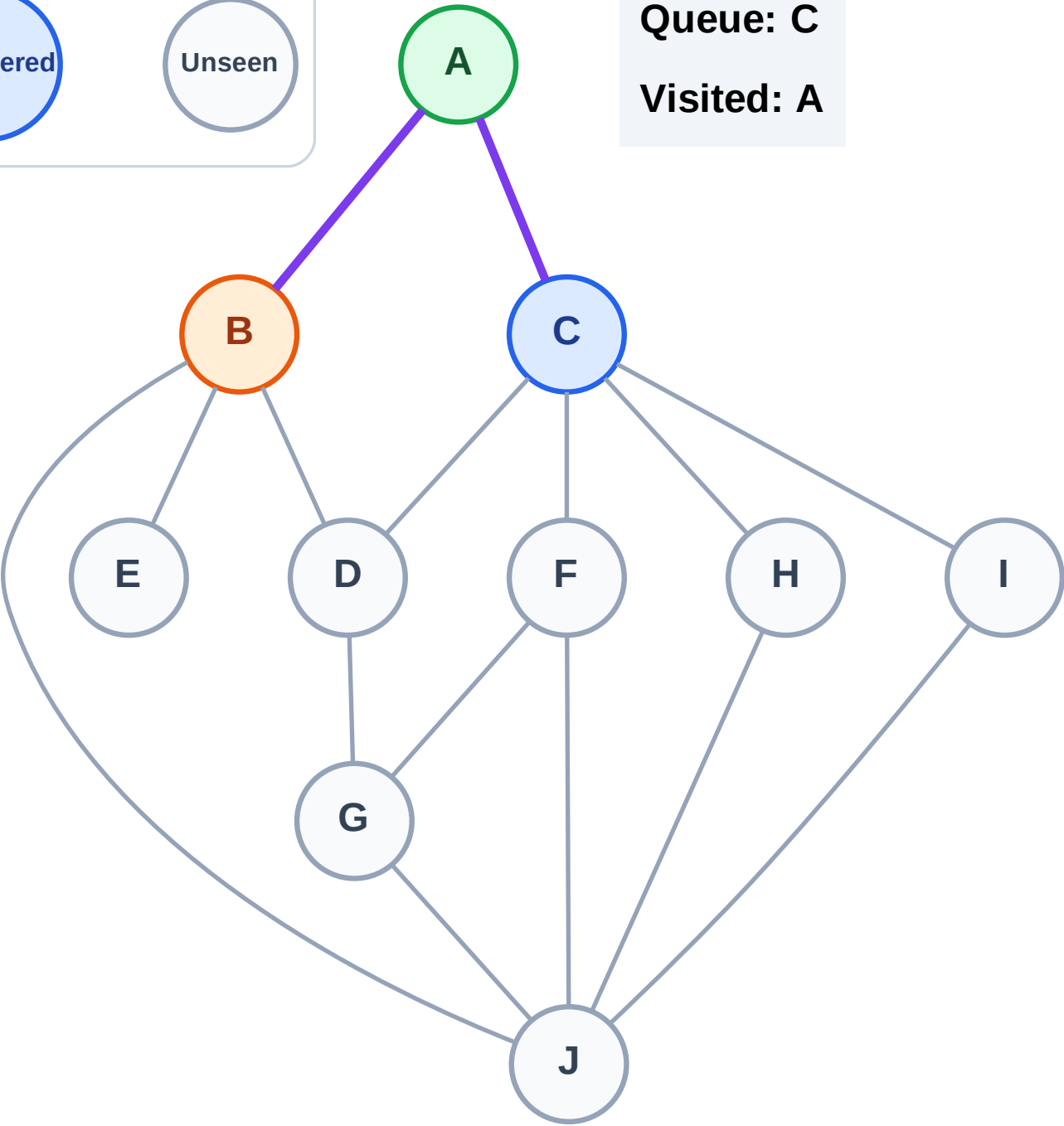
Complete

Processing

Discovered

Unseen

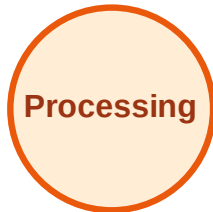
Queue: C
Visited: A



BFS Traversal

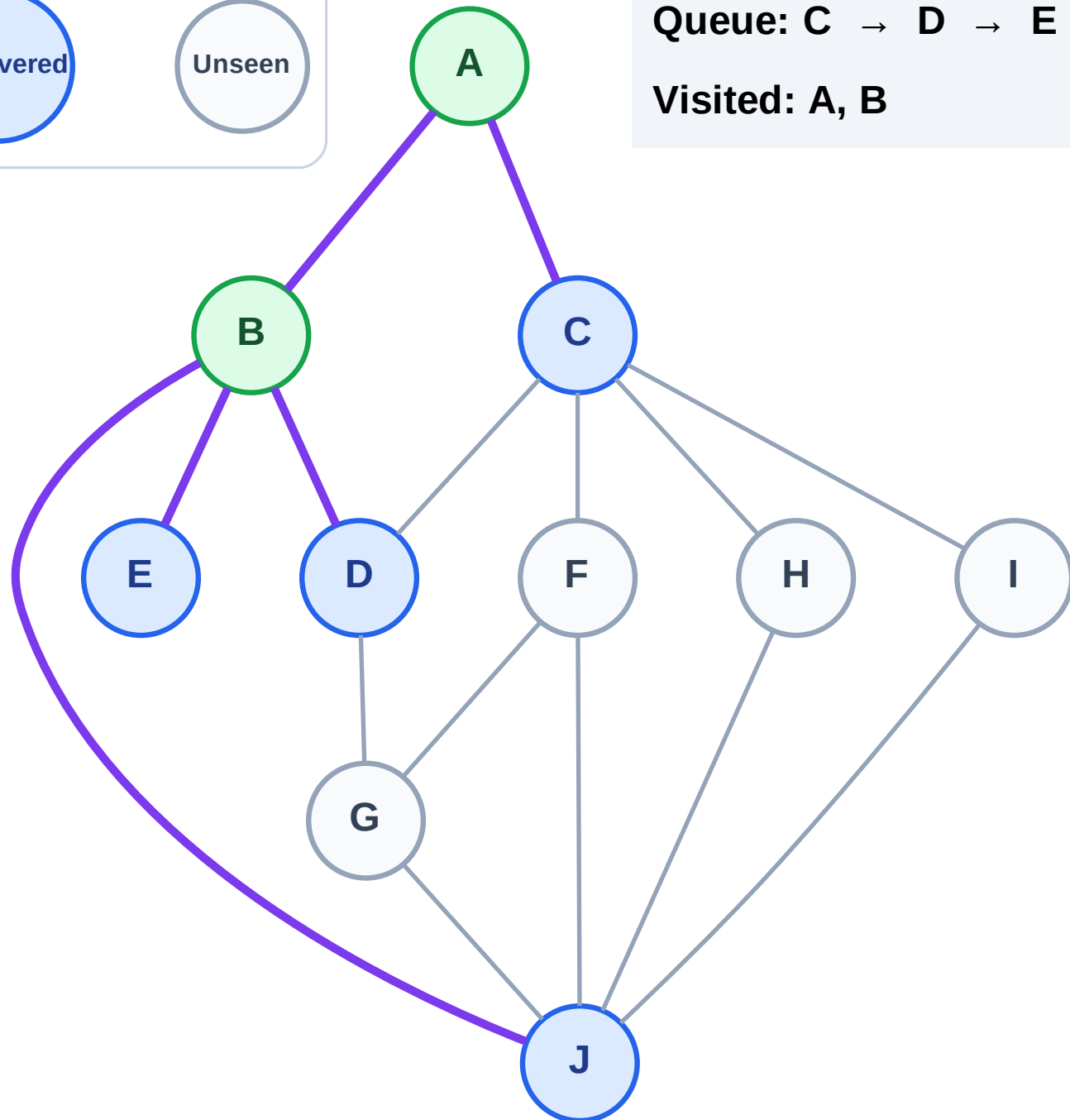
Step 5: Discover D, E, J from B

Legend



Queue: C → D → E → J

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

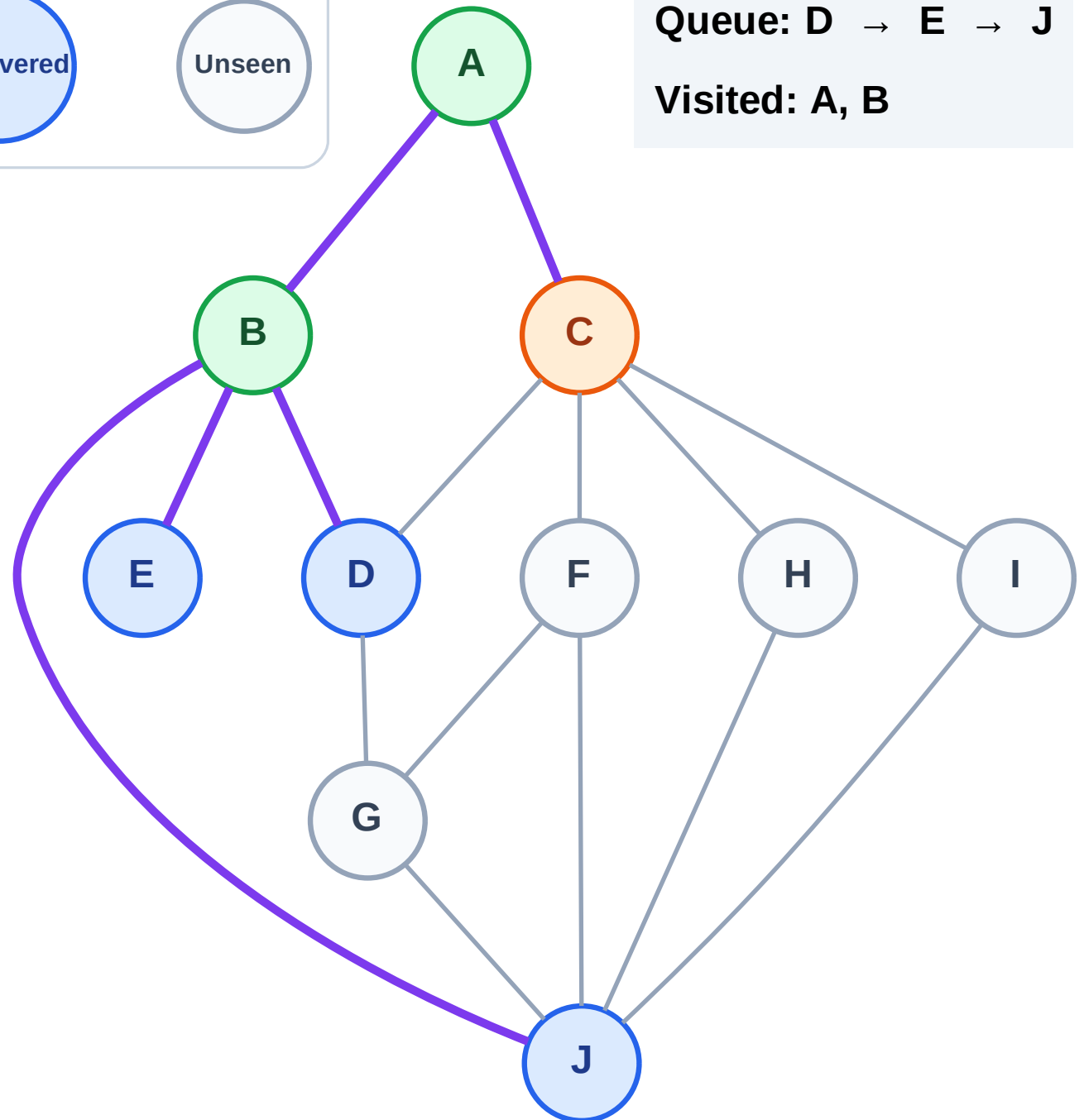
Processing

Discovered

Unseen

Queue: D → E → J

Visited: A, B



BFS Traversal

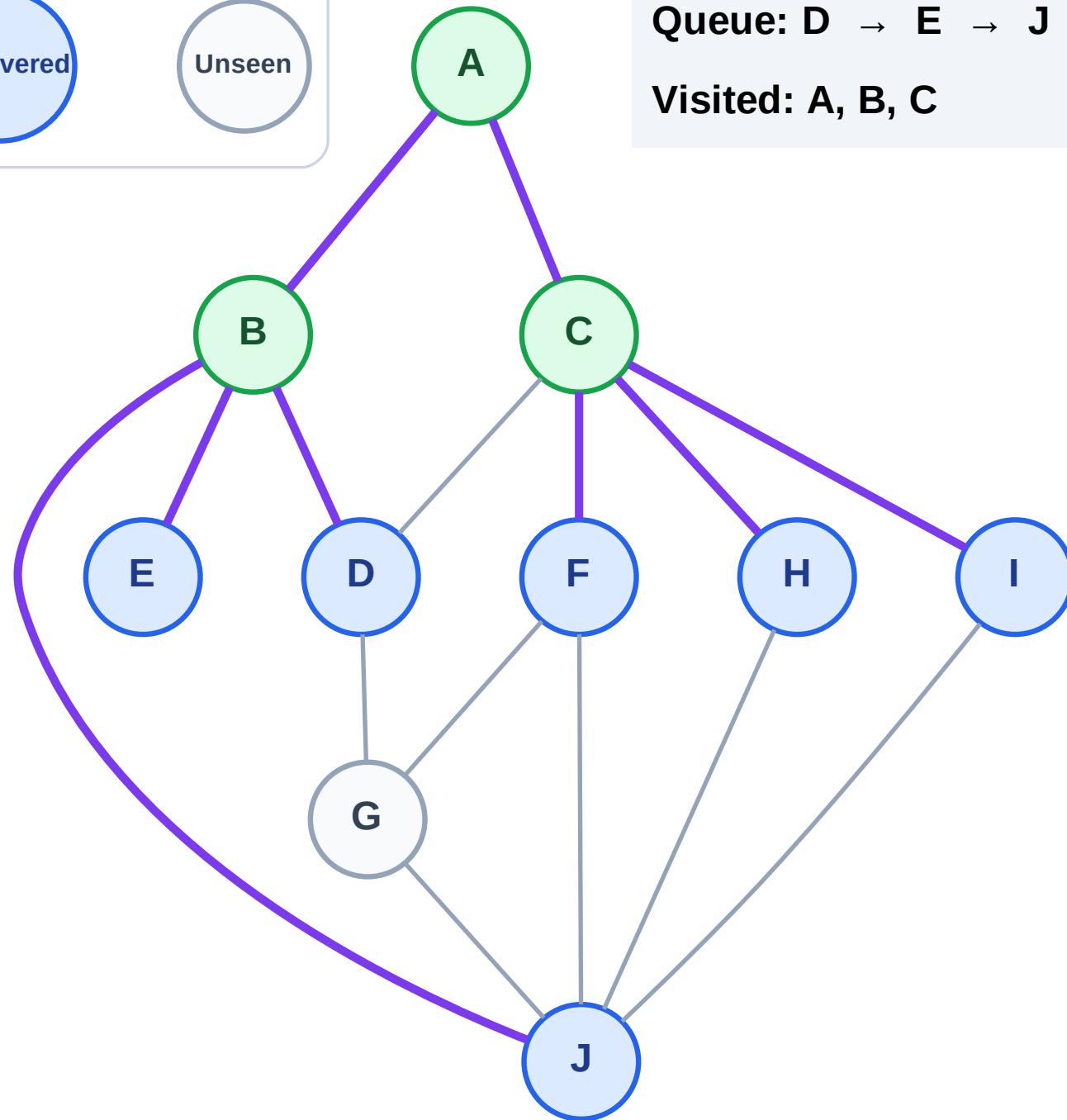
Step 7: Discover F, H, I from C

Legend



Queue: D → E → J → F → H → I

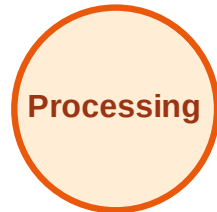
Visited: A, B, C



BFS Traversal

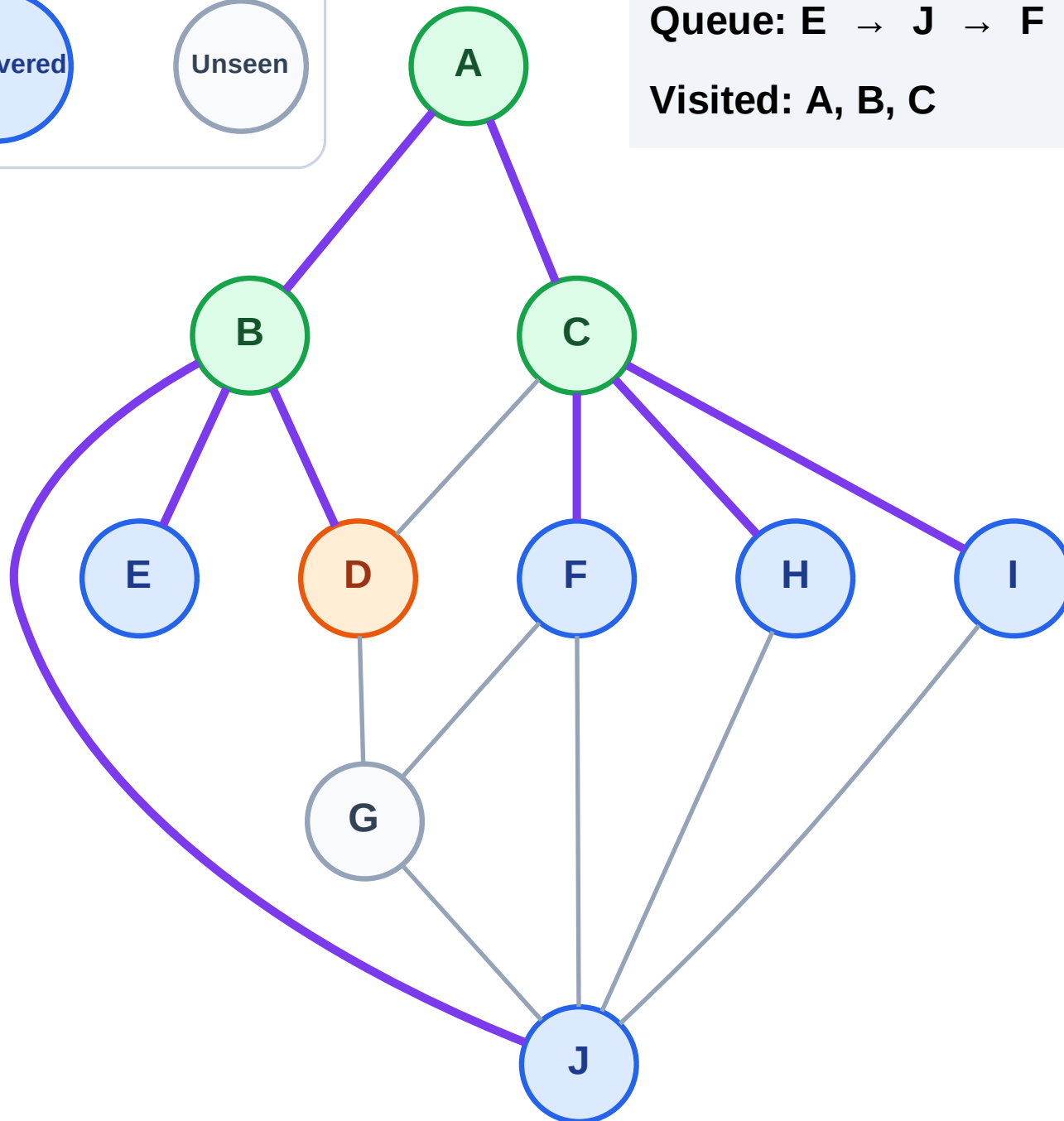
Step 8: Process node D

Legend



Queue: E → J → F → H → I

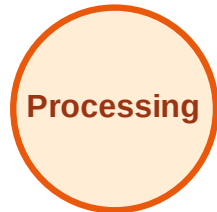
Visited: A, B, C



BFS Traversal

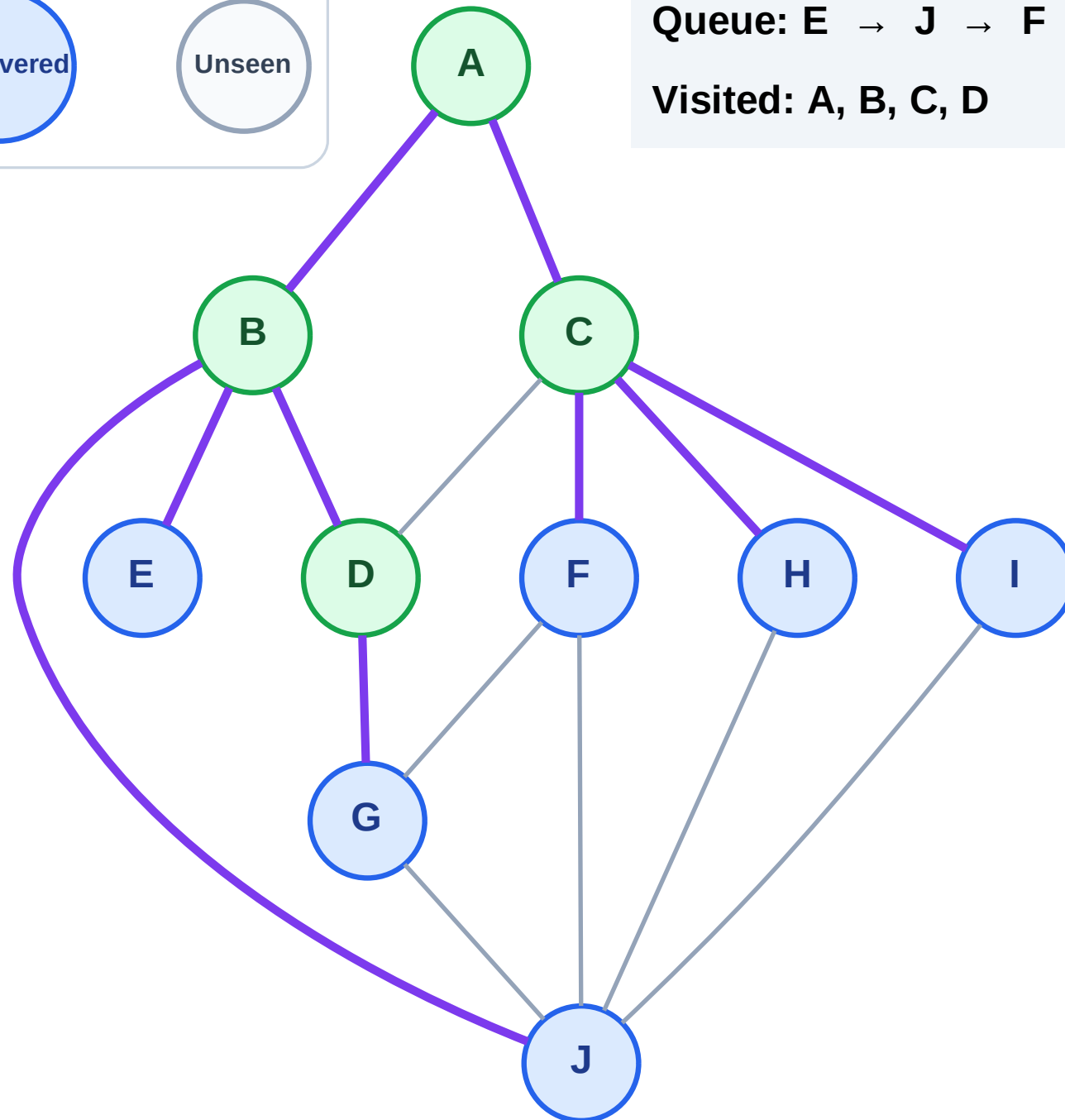
Step 9: Discover G from D

Legend



Queue: E → J → F → H → I → G

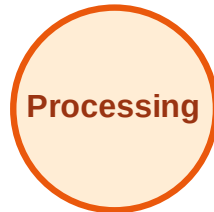
Visited: A, B, C, D



BFS Traversal

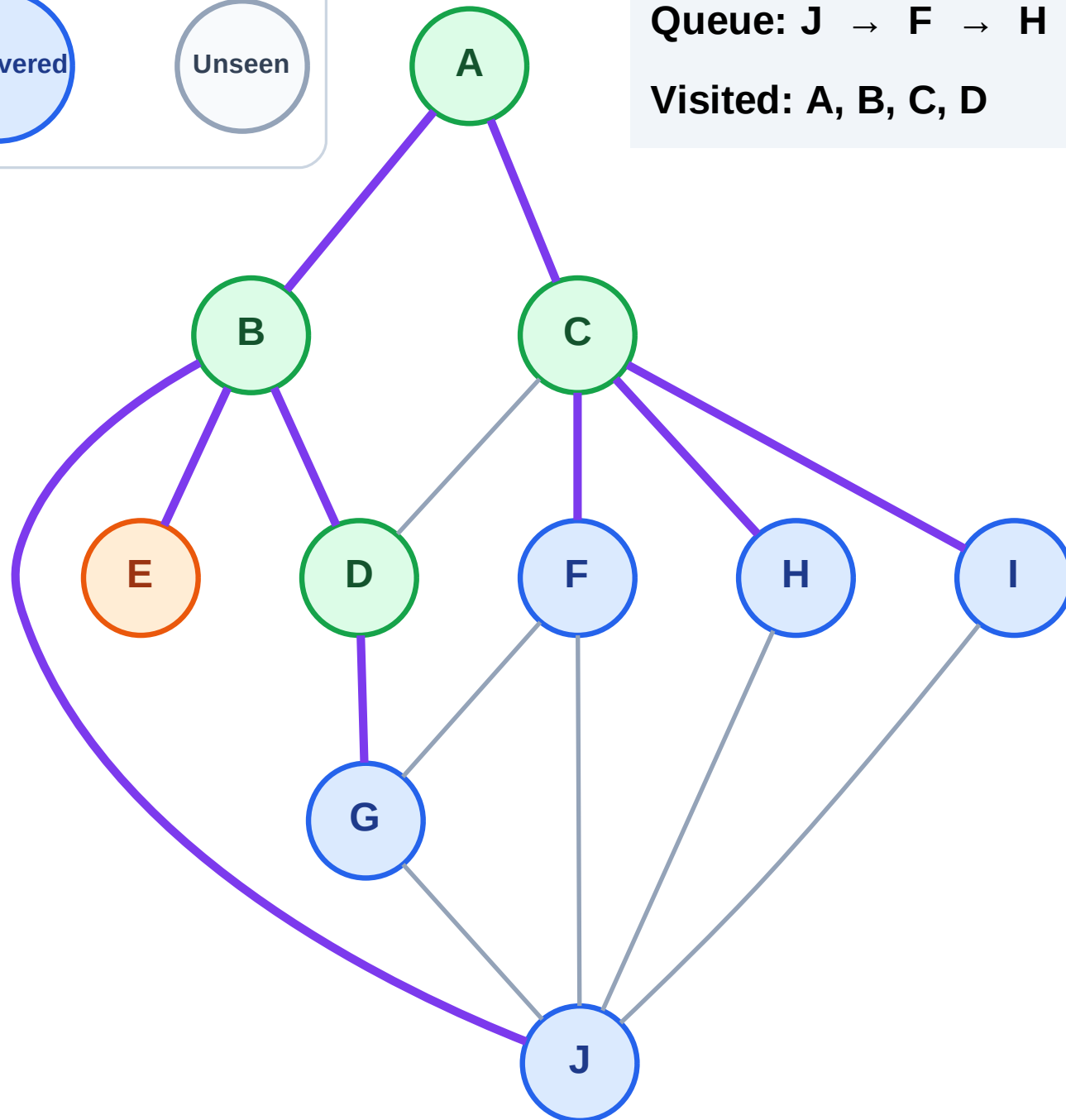
Step 10: Process node E

Legend



Queue: J → F → H → I → G

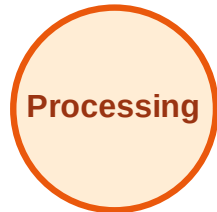
Visited: A, B, C, D



BFS Traversal

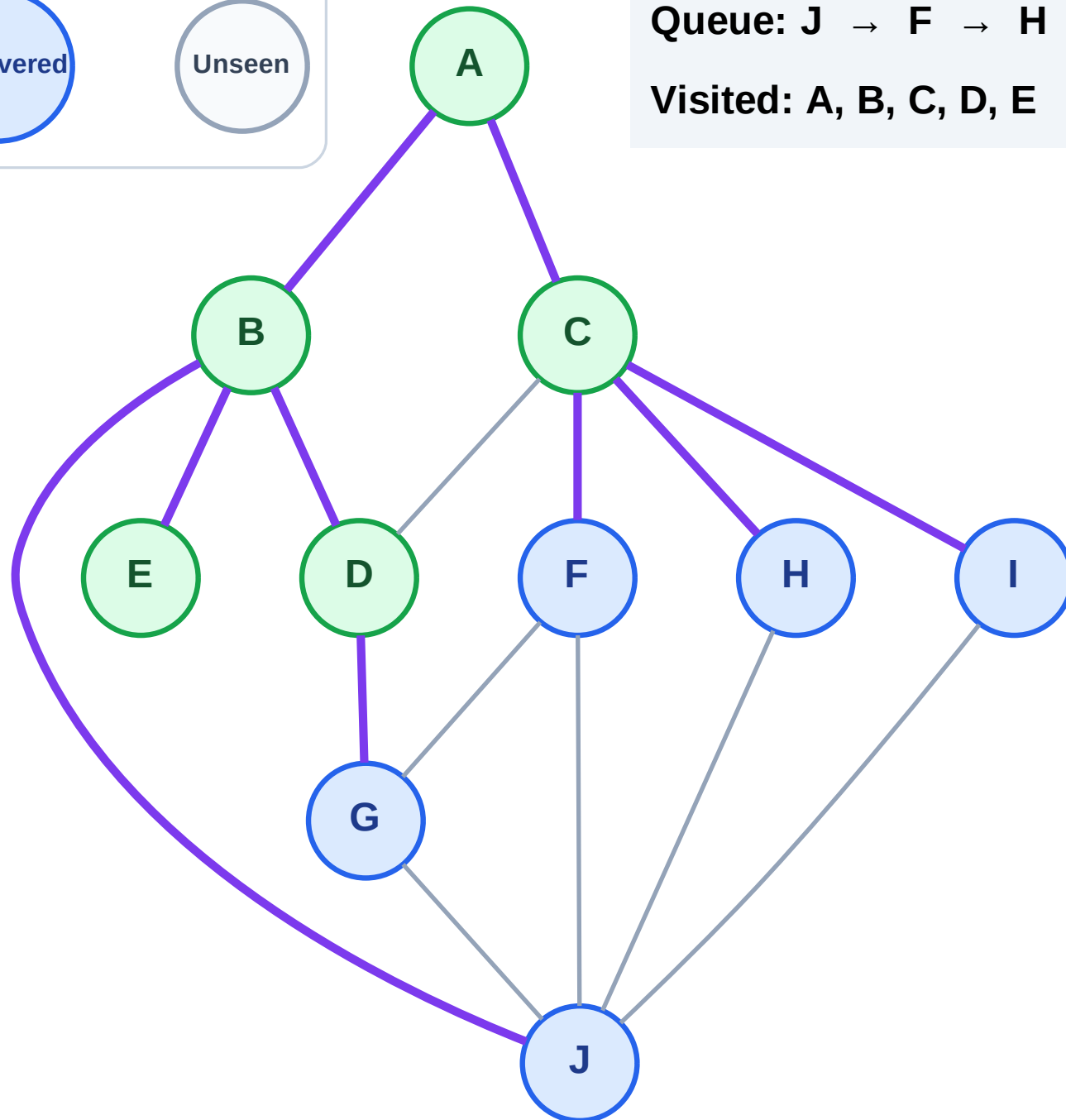
Step 11: No new neighbors from E

Legend



Queue: J → F → H → I → G

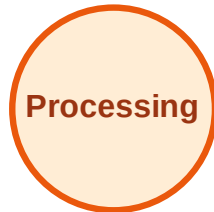
Visited: A, B, C, D, E



BFS Traversal

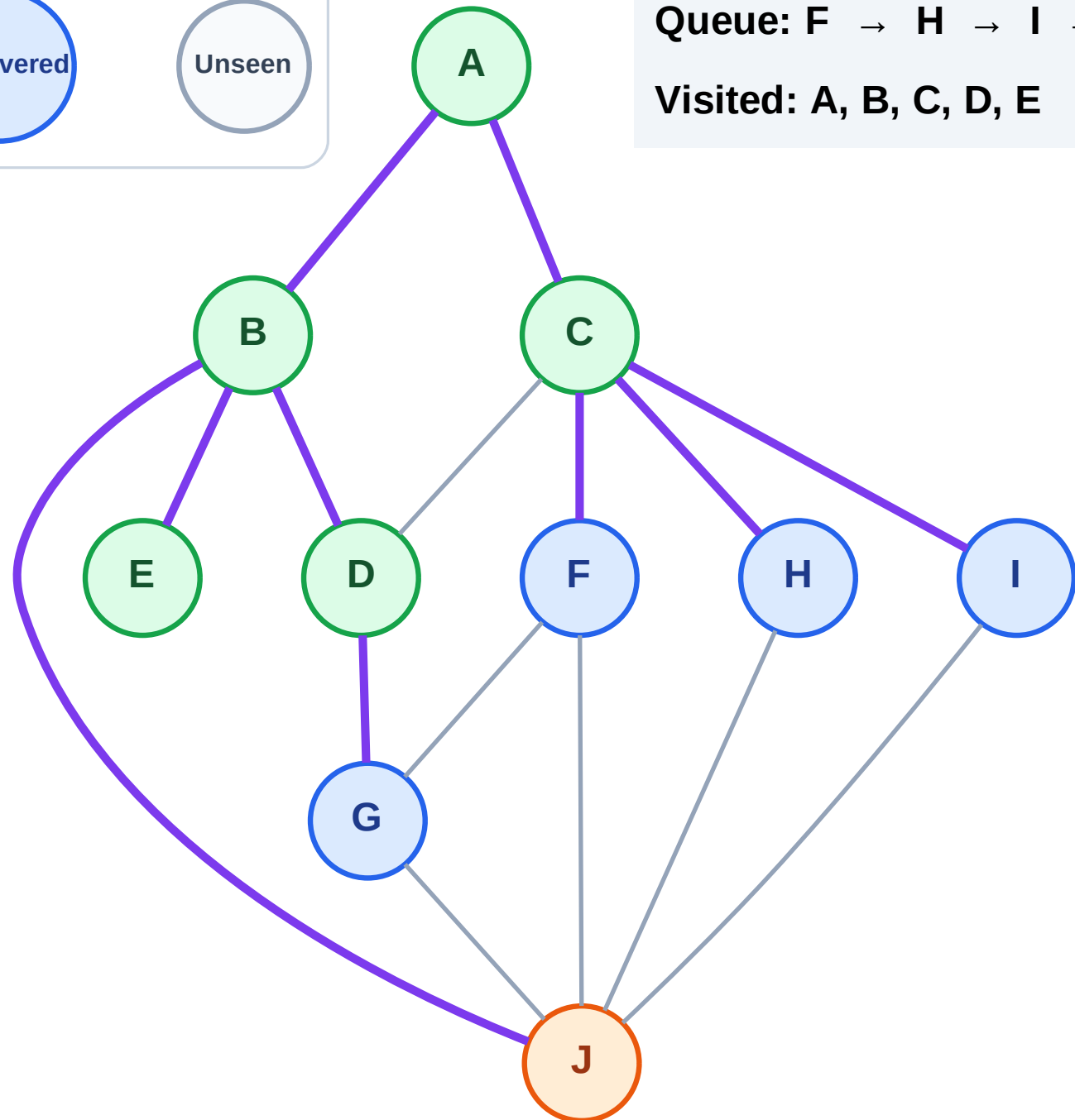
Step 12: Process node J

Legend



Queue: F → H → I → G

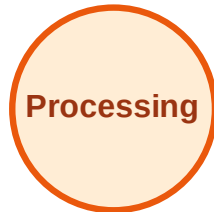
Visited: A, B, C, D, E



BFS Traversal

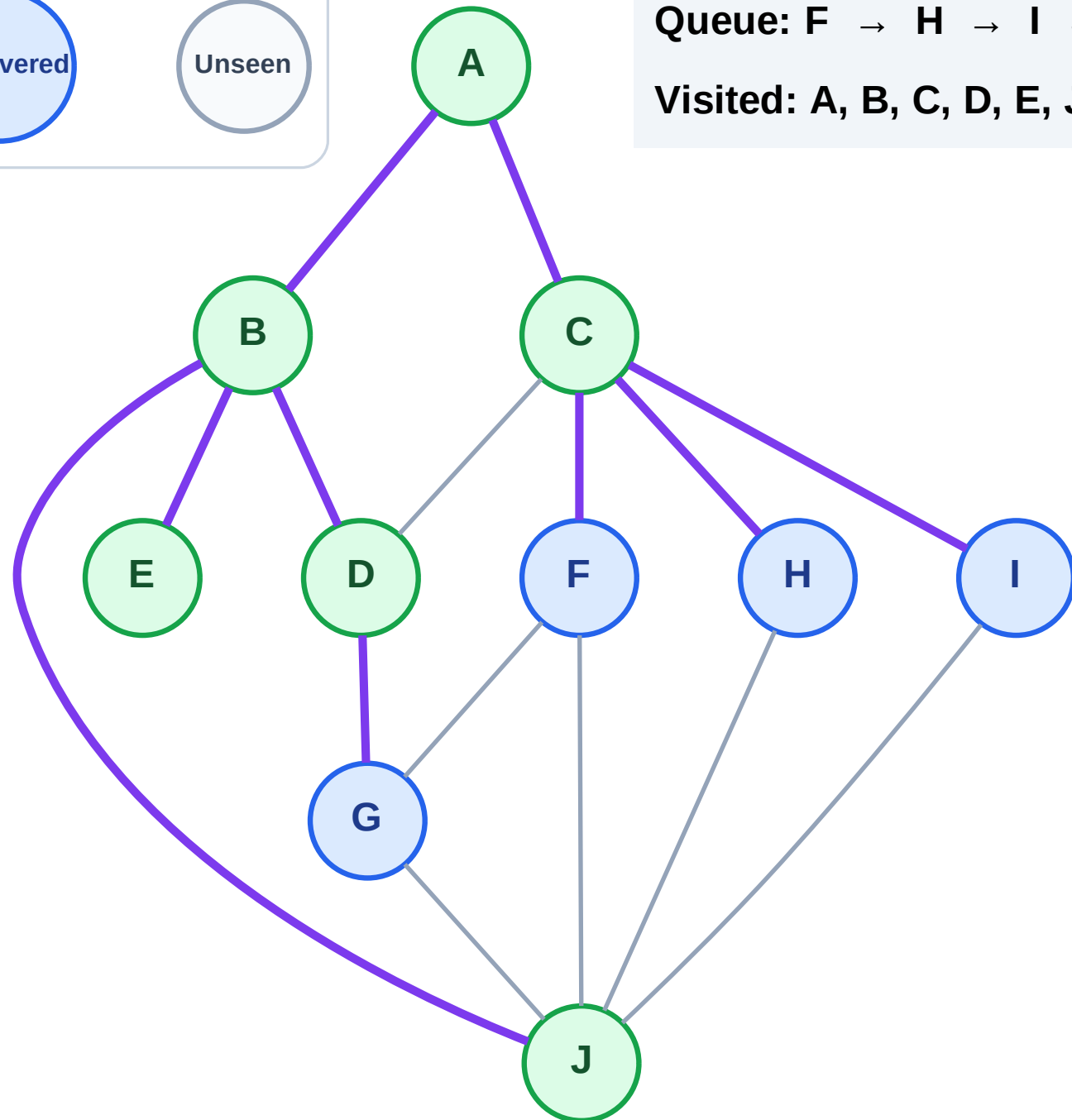
Step 13: No new neighbors from J

Legend



Queue: F → H → I → G

Visited: A, B, C, D, E, J



BFS Traversal

Step 14: Process node F

Legend

Complete

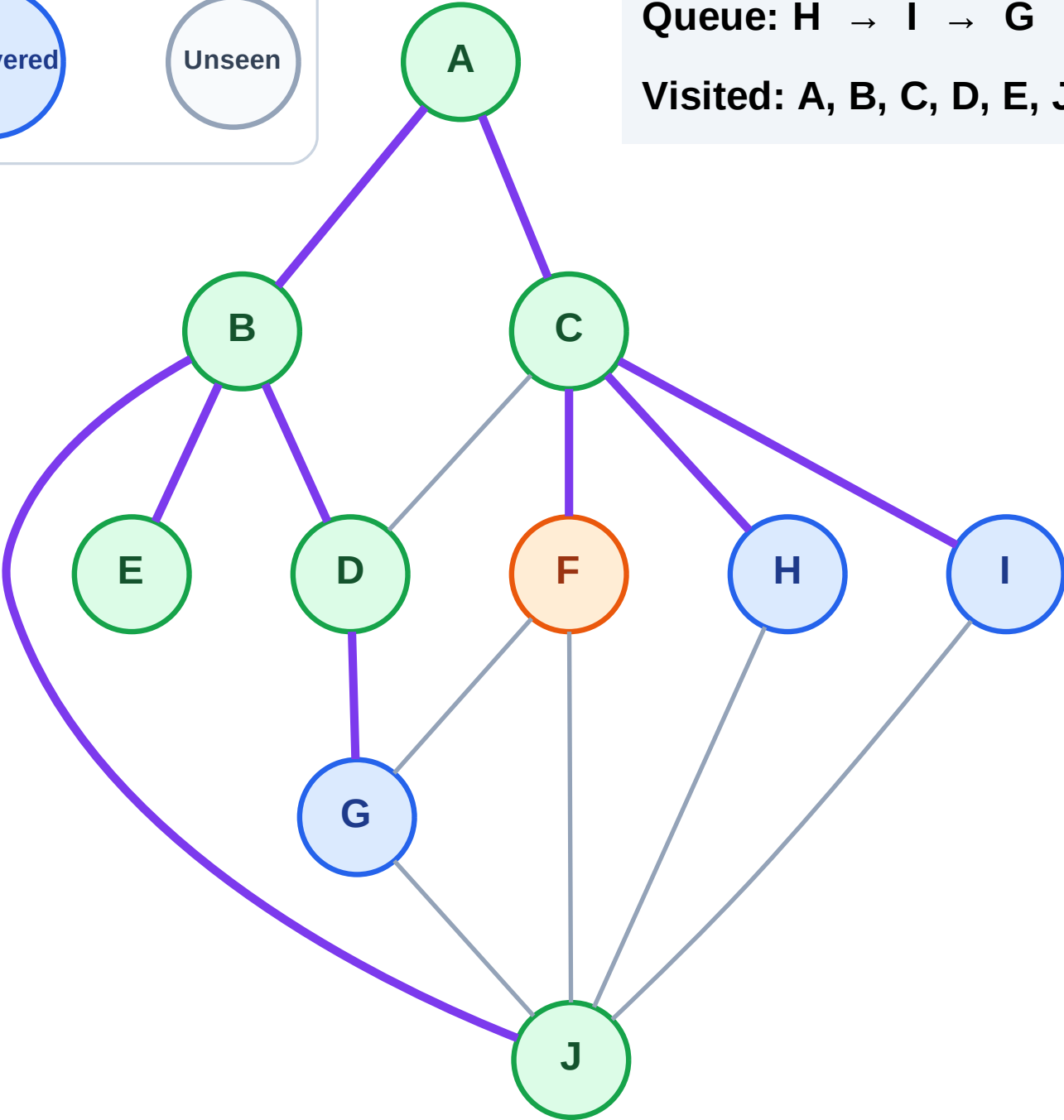
Processing

Discovered

Unseen

Queue: H → I → G

Visited: A, B, C, D, E, J



BFS Traversal

Step 15: No new neighbors from F

Legend

Complete

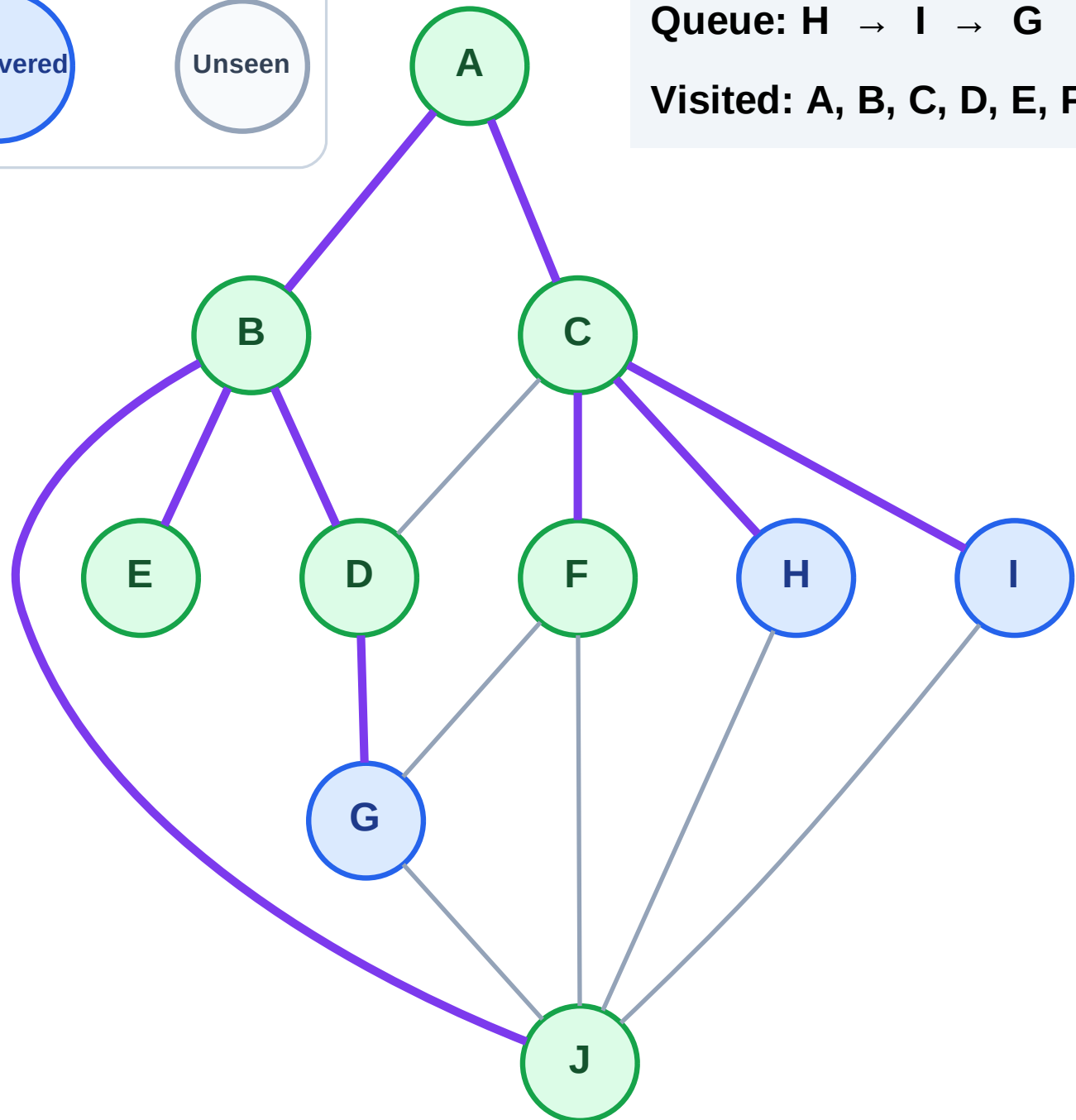
Processing

Discovered

Unseen

Queue: H → I → G

Visited: A, B, C, D, E, F, J



BFS Traversal

Step 16: Process node H

Legend

Complete

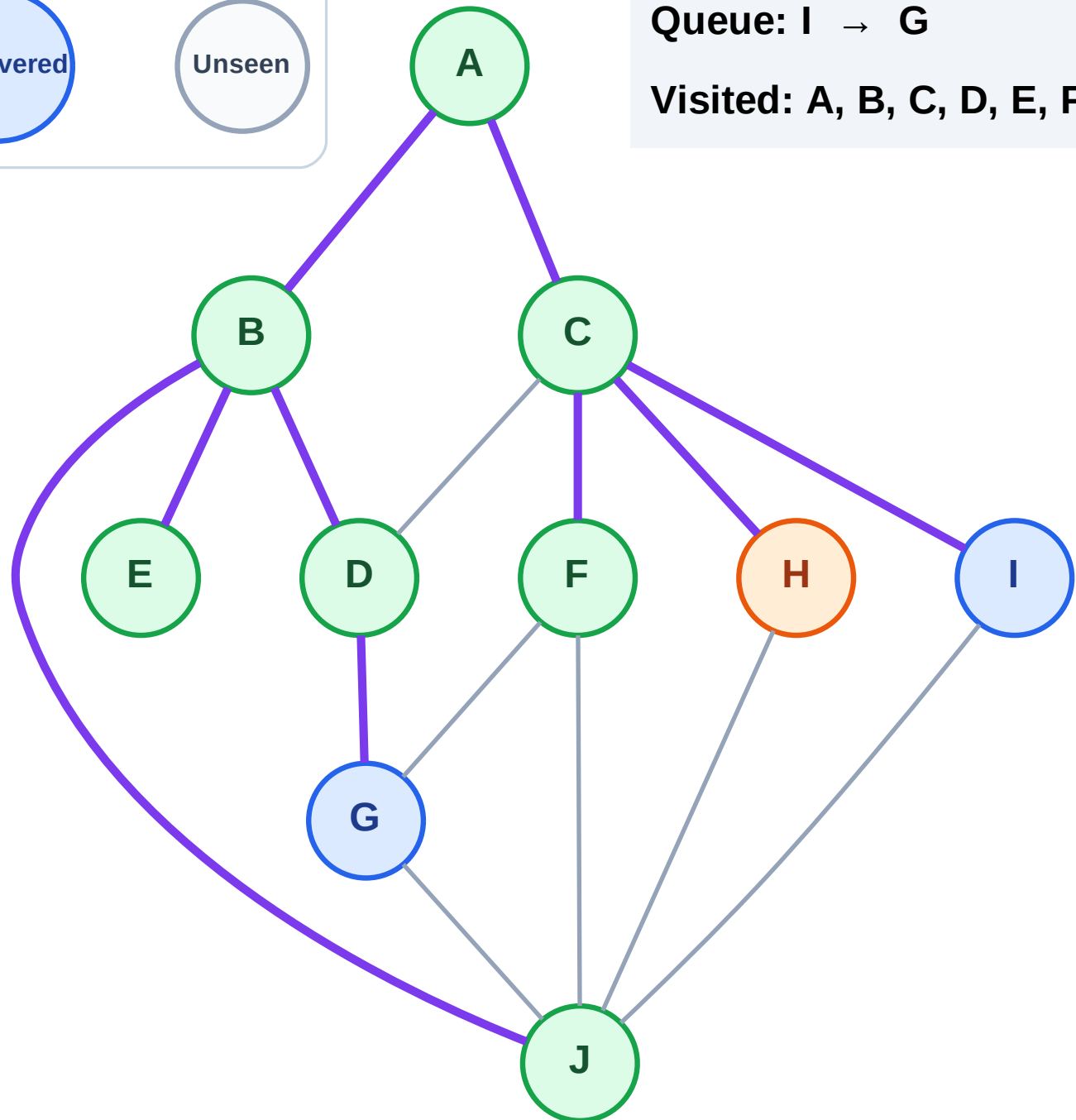
Processing

Discovered

Unseen

Queue: I → G

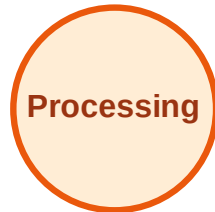
Visited: A, B, C, D, E, F, J



BFS Traversal

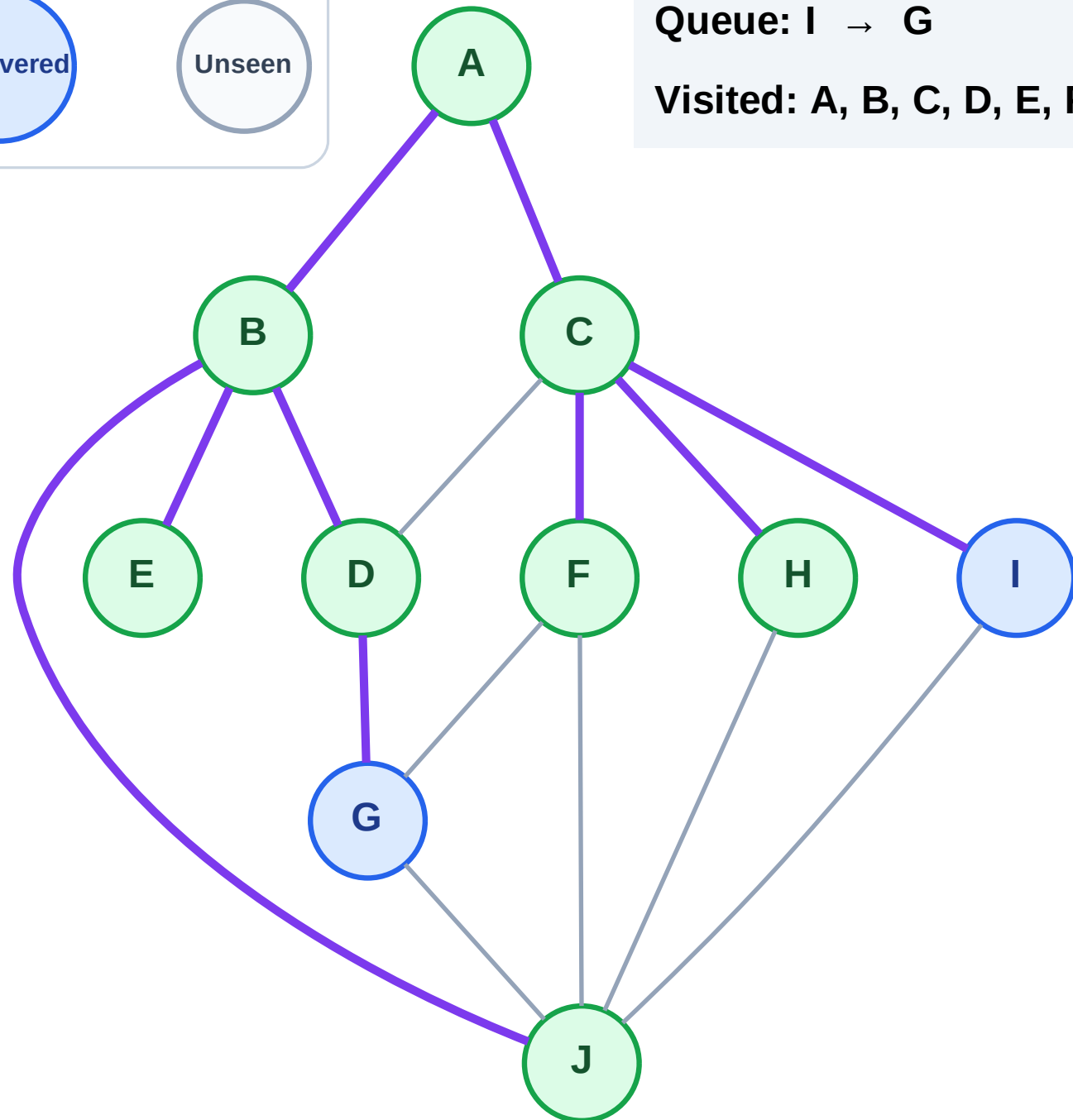
Step 17: No new neighbors from H

Legend



Queue: I → G

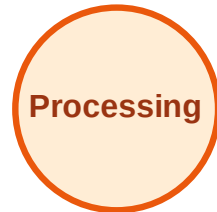
Visited: A, B, C, D, E, F, H, J



BFS Traversal

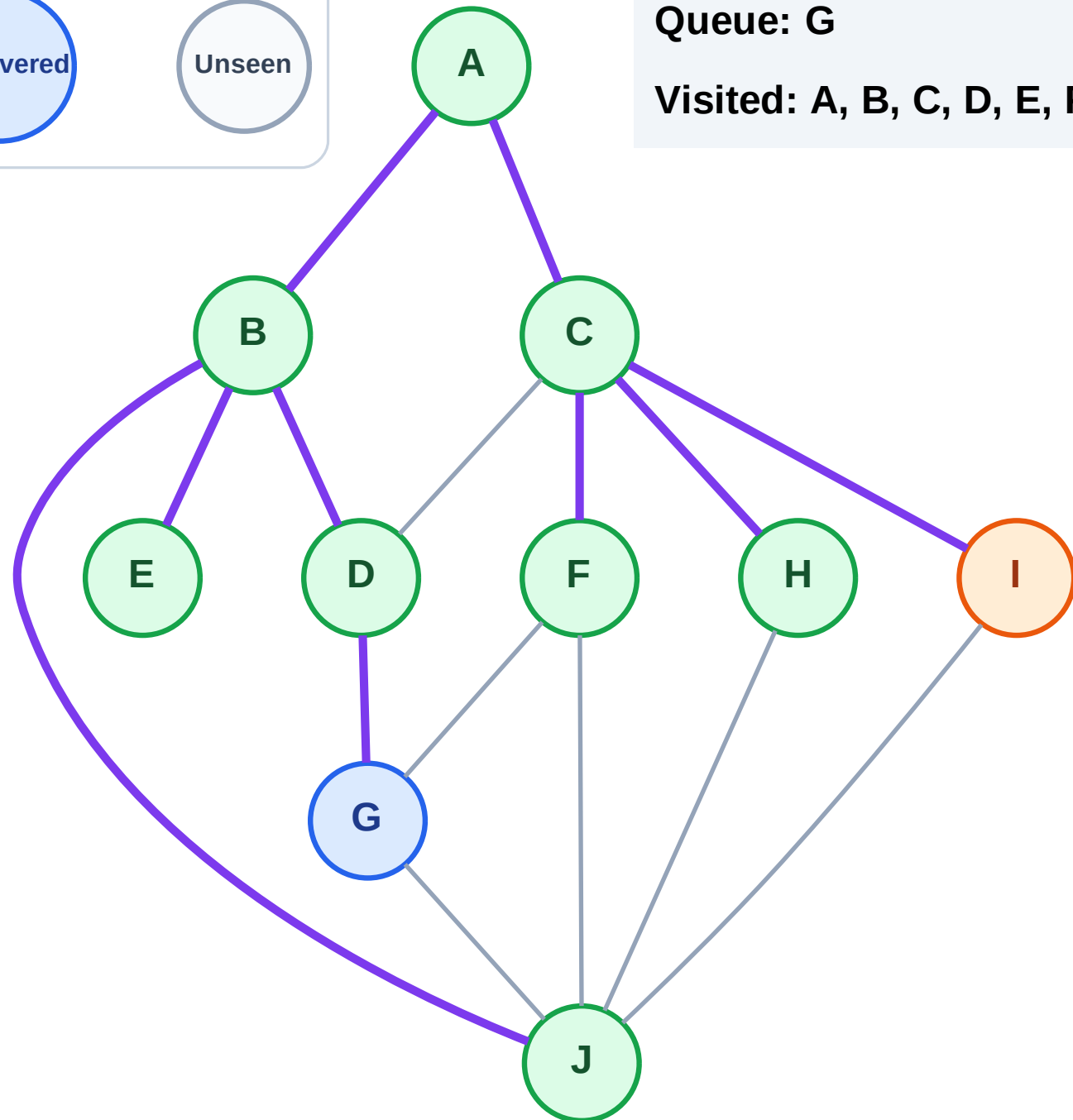
Step 18: Process node I

Legend



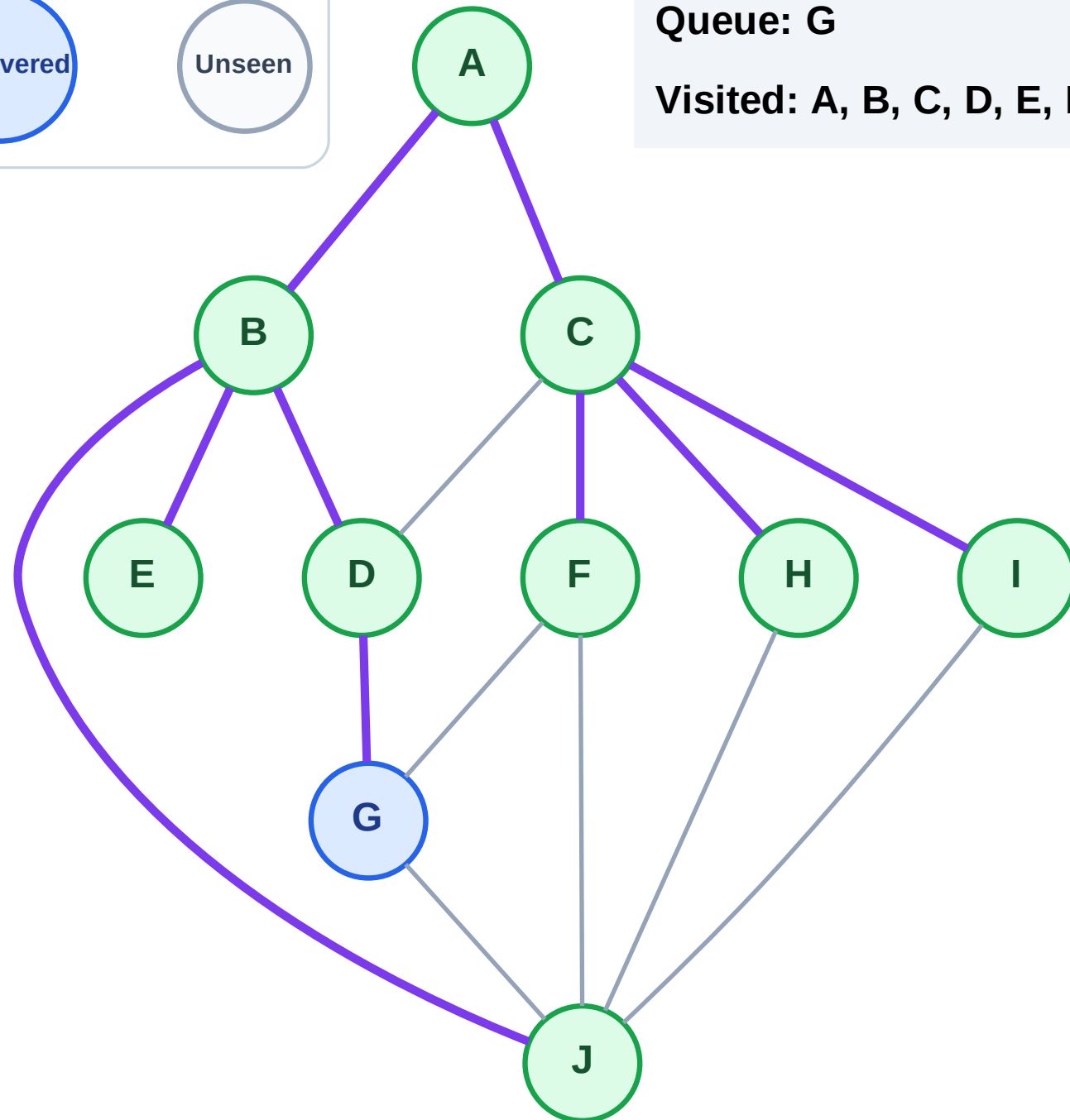
Queue: G

Visited: A, B, C, D, E, F, H, J



Step 19: No new neighbors from I

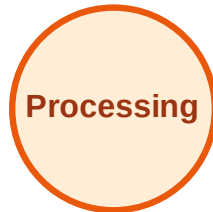
Step 19: No new neighbors from I



BFS Traversal

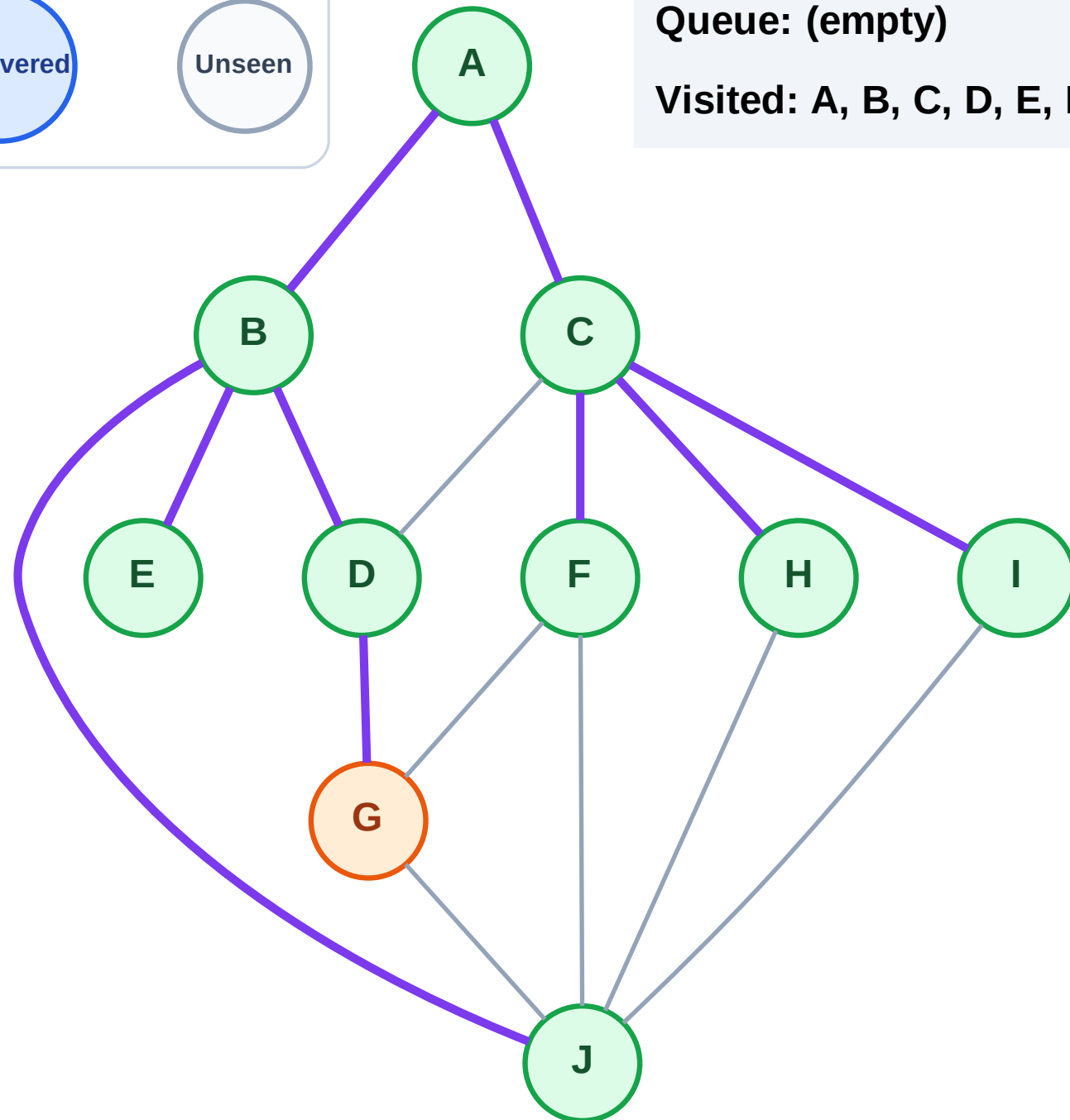
Step 20: Process node G

Legend



Queue: (empty)

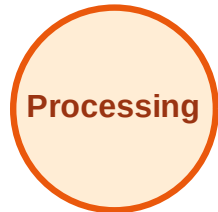
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

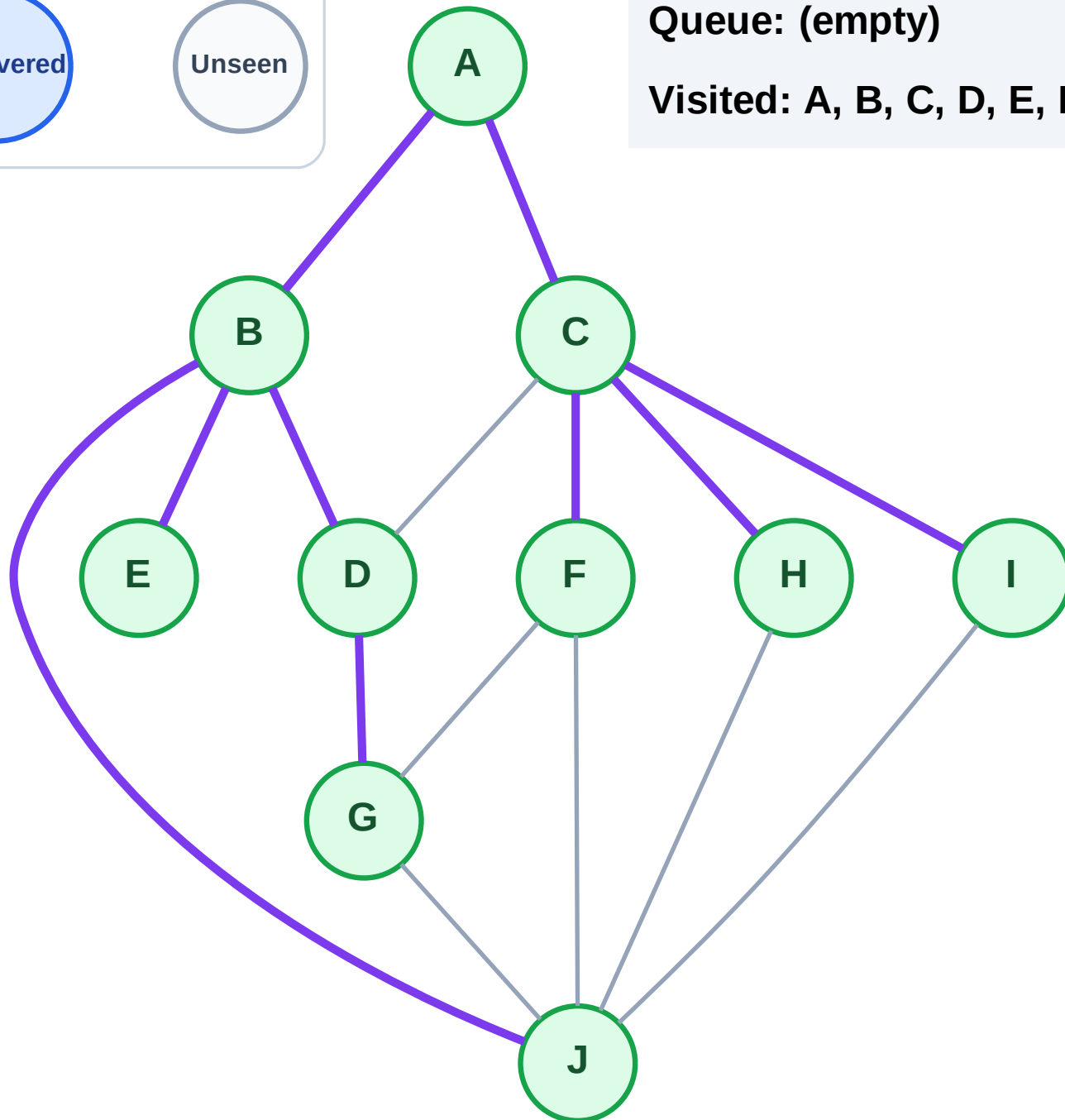
Step 21: No new neighbors from G

Legend



Queue: (empty)

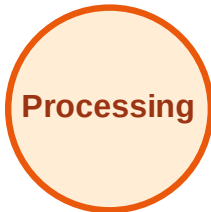
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

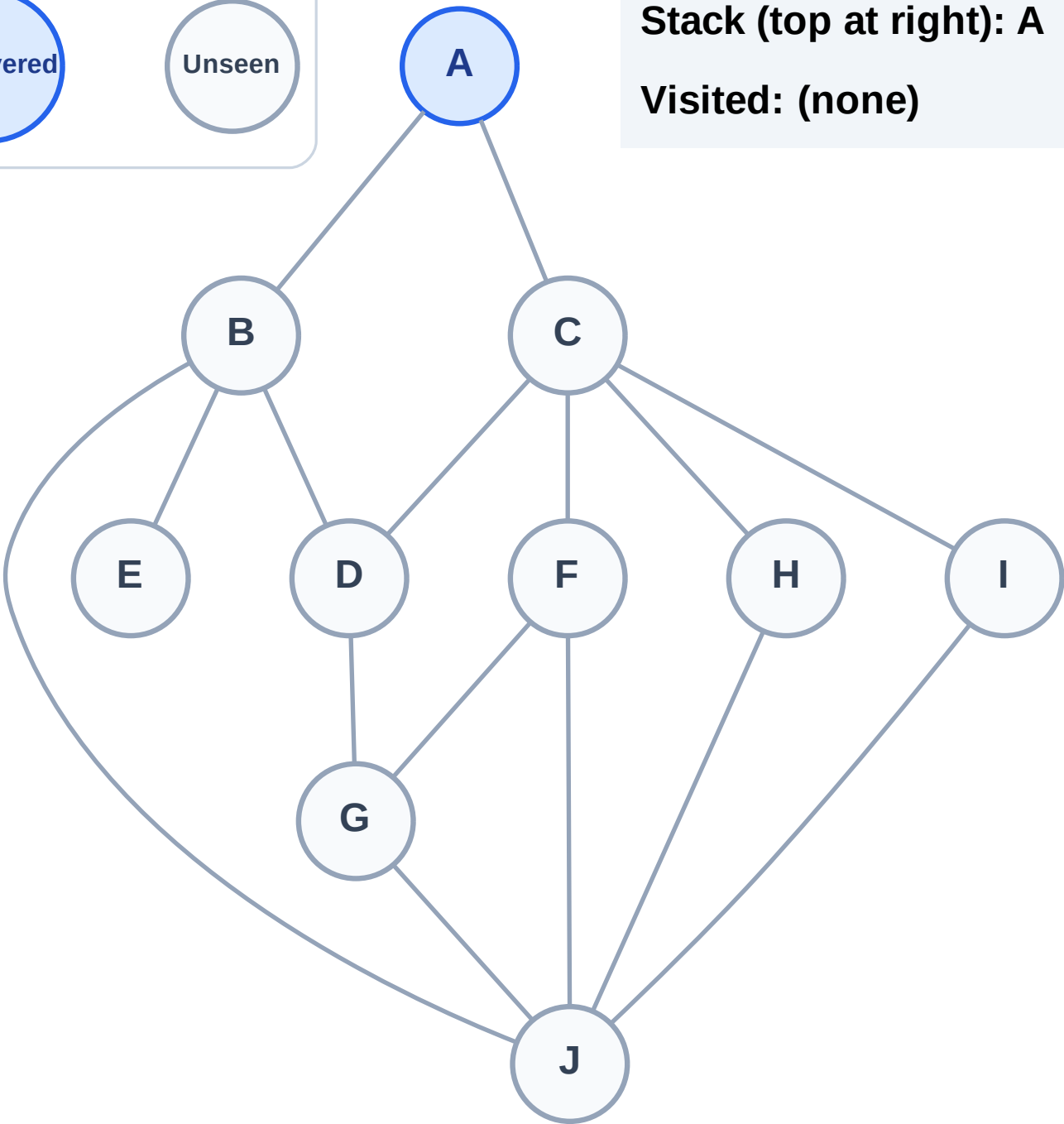
Step 1: Start at A

Legend



Stack (top at right): A

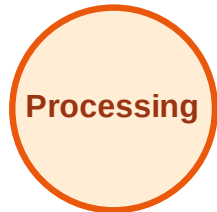
Visited: (none)



DFS Traversal

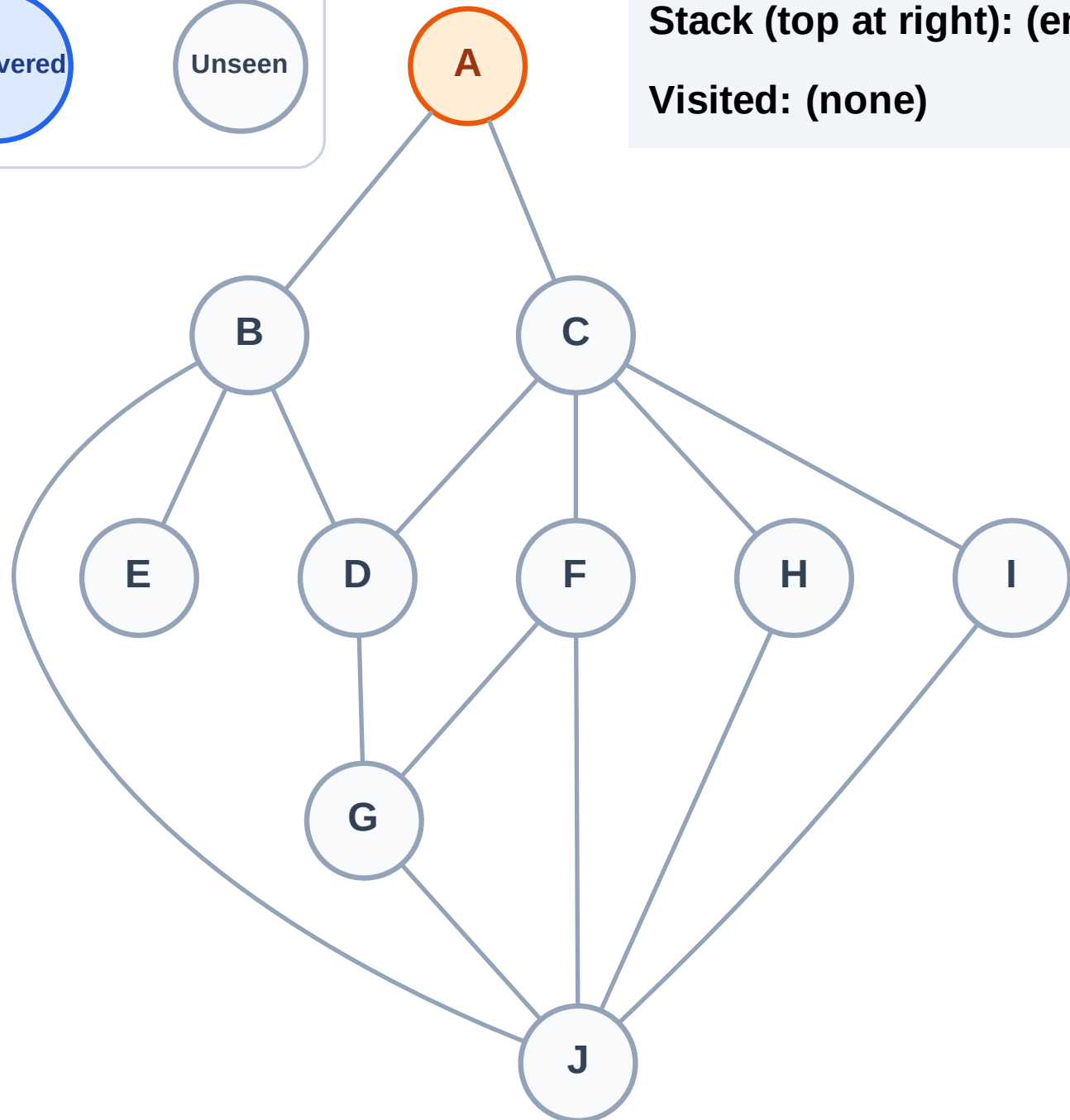
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

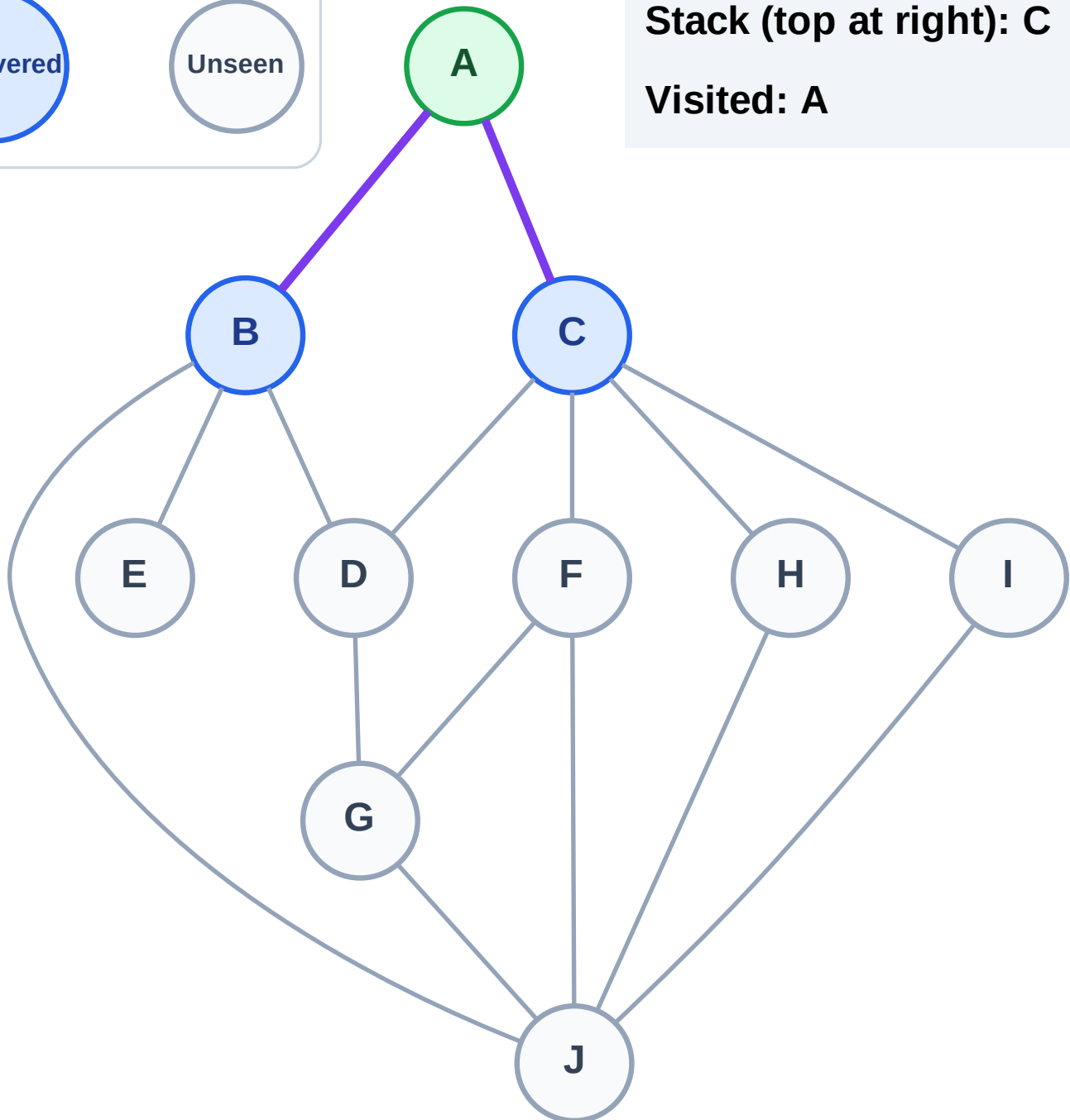
Step 3: Discover C, B from A

Legend



Stack (top at right): C | B

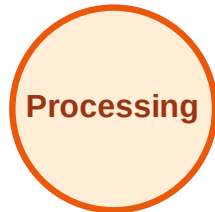
Visited: A



DFS Traversal

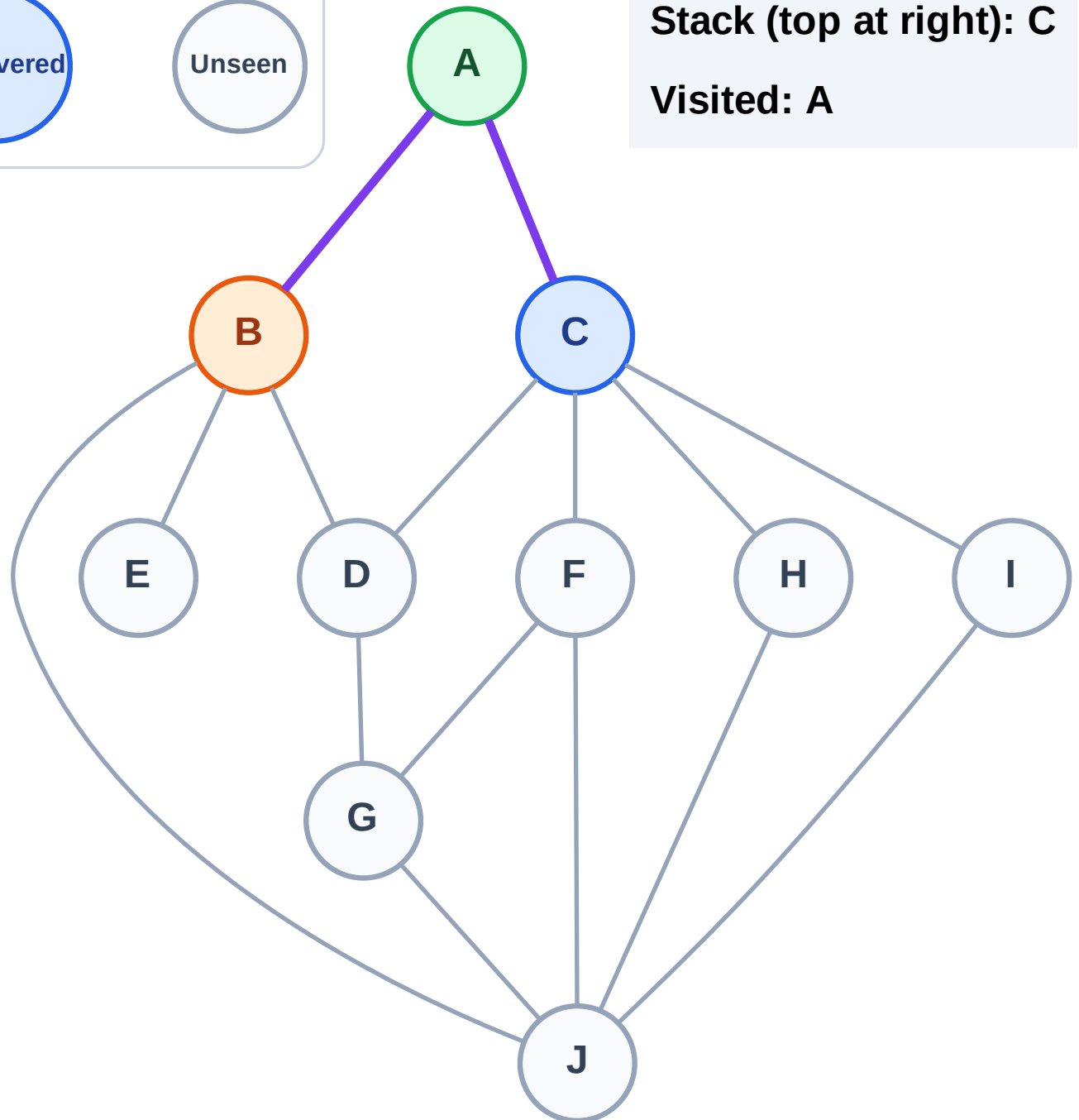
Step 4: Process node B

Legend



Stack (top at right): C

Visited: A



DFS Traversal

Step 5: Discover J, E, D from B

Legend

Complete

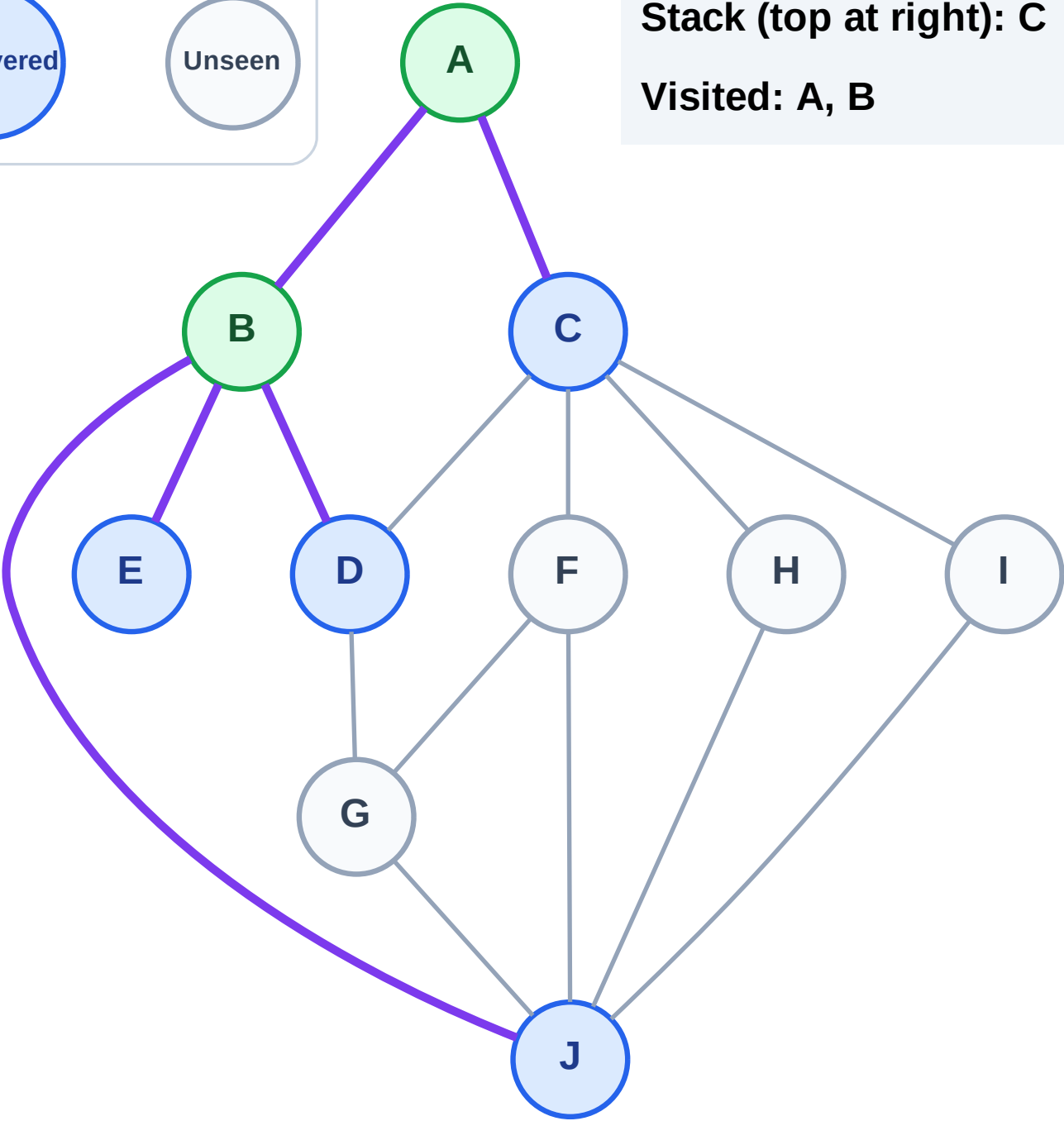
Processing

Discovered

Unseen

Stack (top at right): C | J | E | D

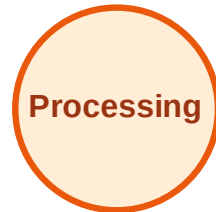
Visited: A, B



DFS Traversal

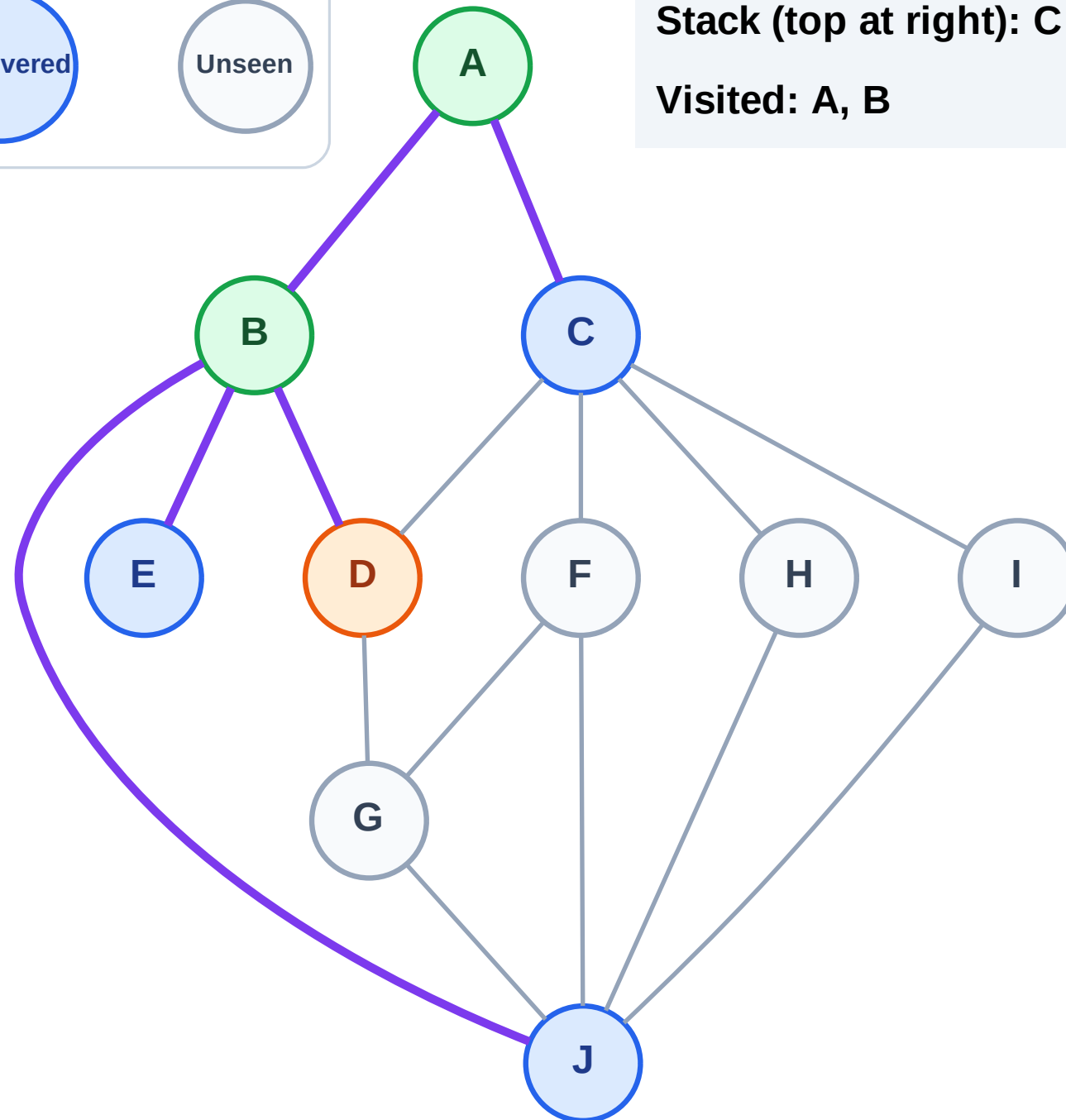
Step 6: Process node D

Legend



Stack (top at right): C | J | E

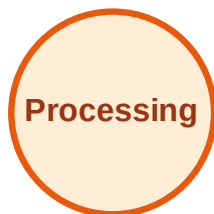
Visited: A, B



DFS Traversal

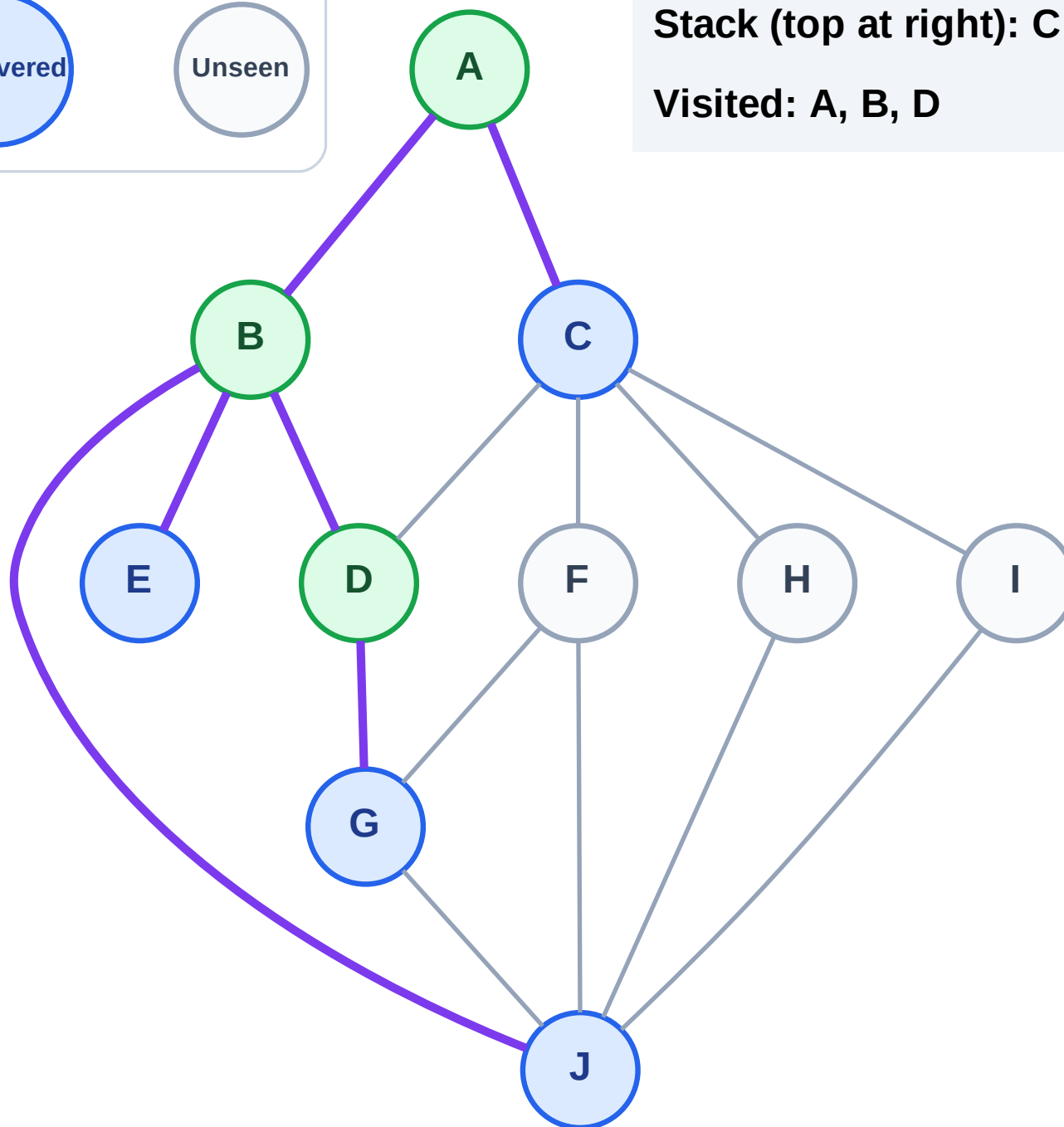
Step 7: Discover G from D

Legend



Stack (top at right): C | J | E | G

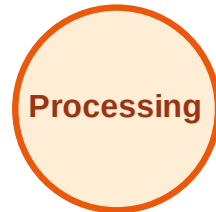
Visited: A, B, D



DFS Traversal

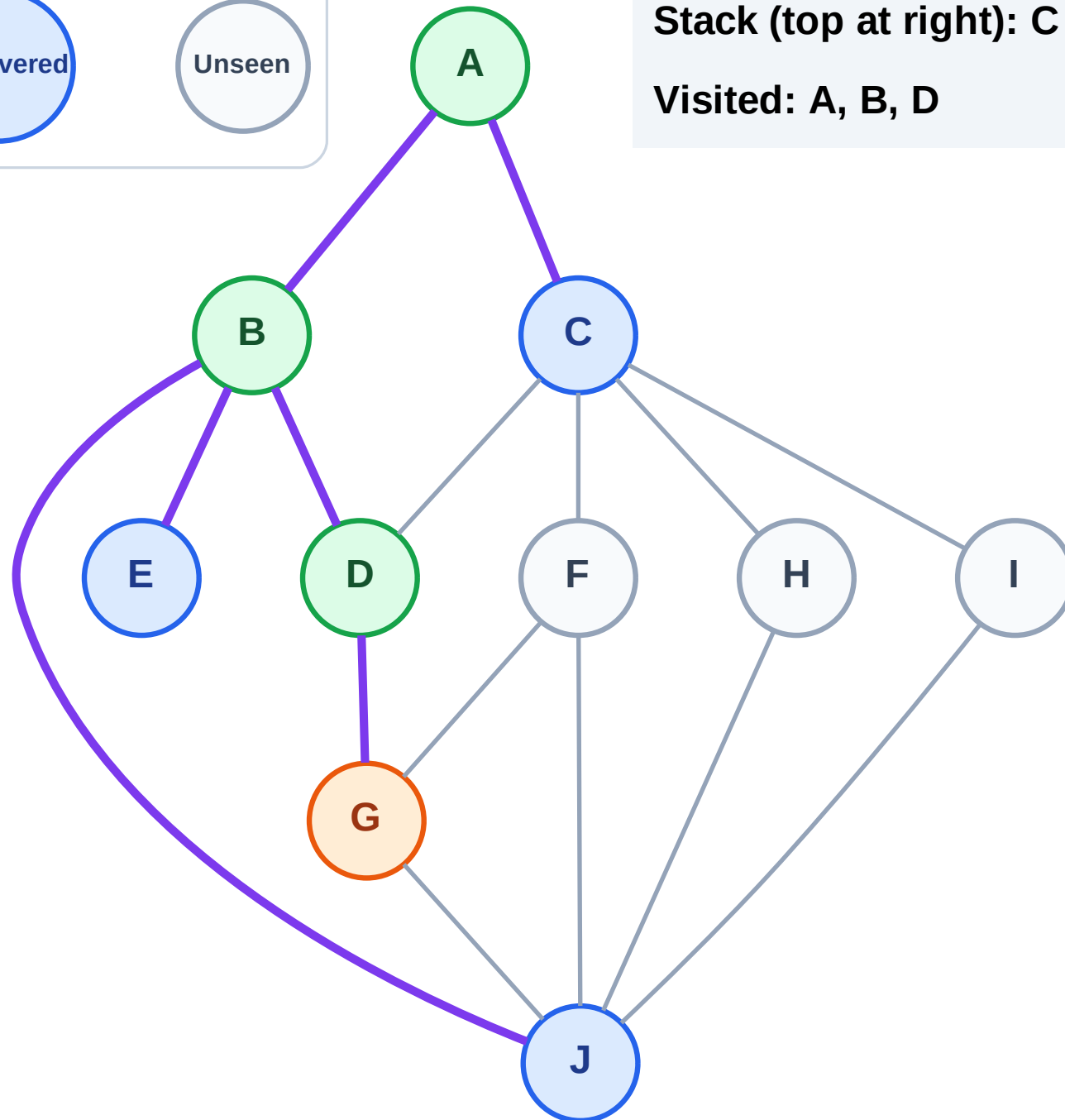
Step 8: Process node G

Legend



Stack (top at right): C | J | E

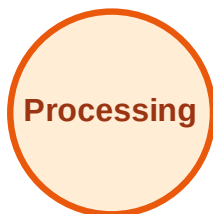
Visited: A, B, D



DFS Traversal

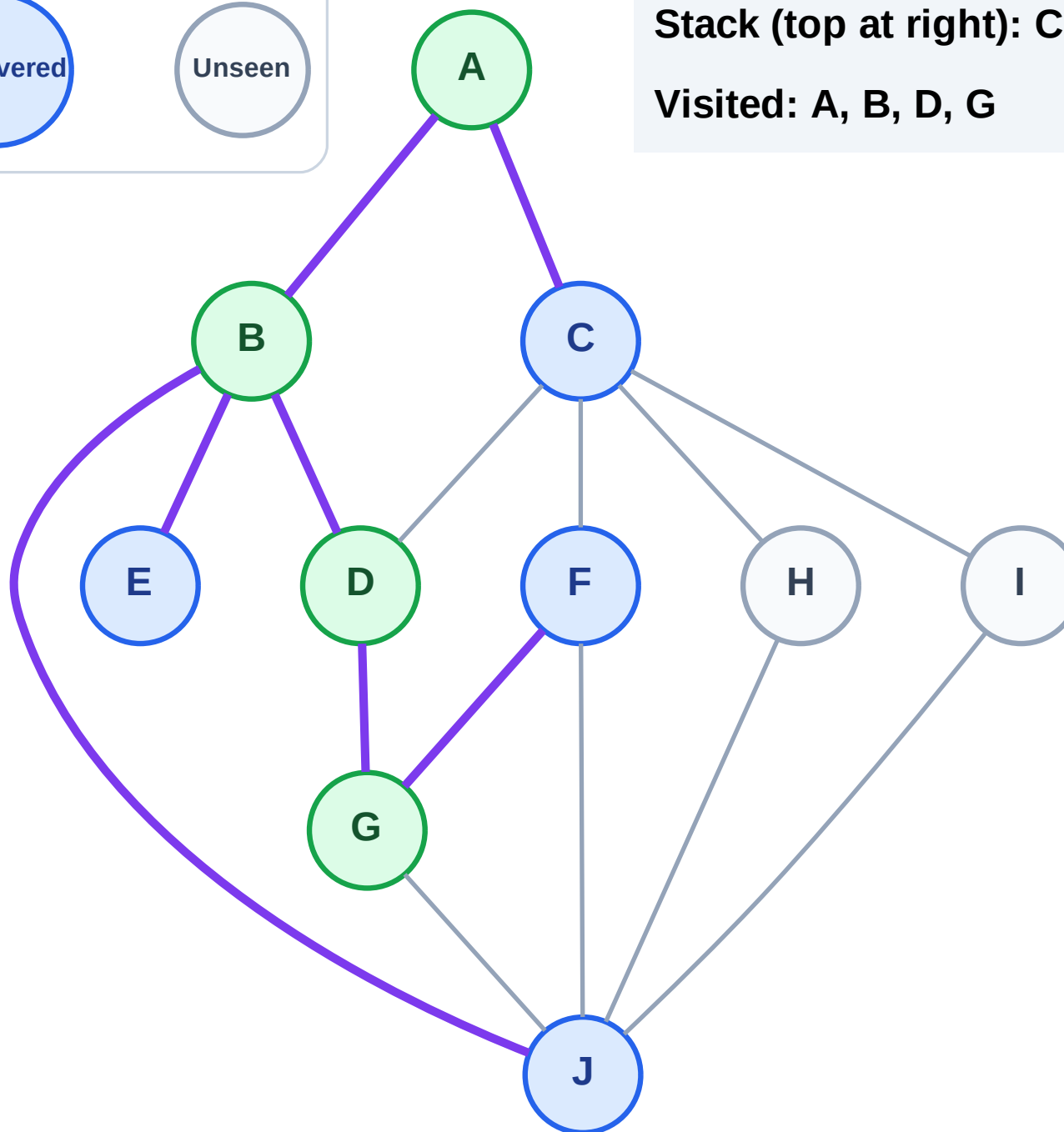
Step 9: Discover F from G

Legend



Stack (top at right): C | J | E | F

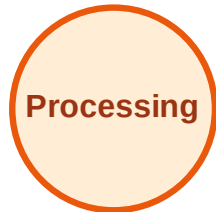
Visited: A, B, D, G



DFS Traversal

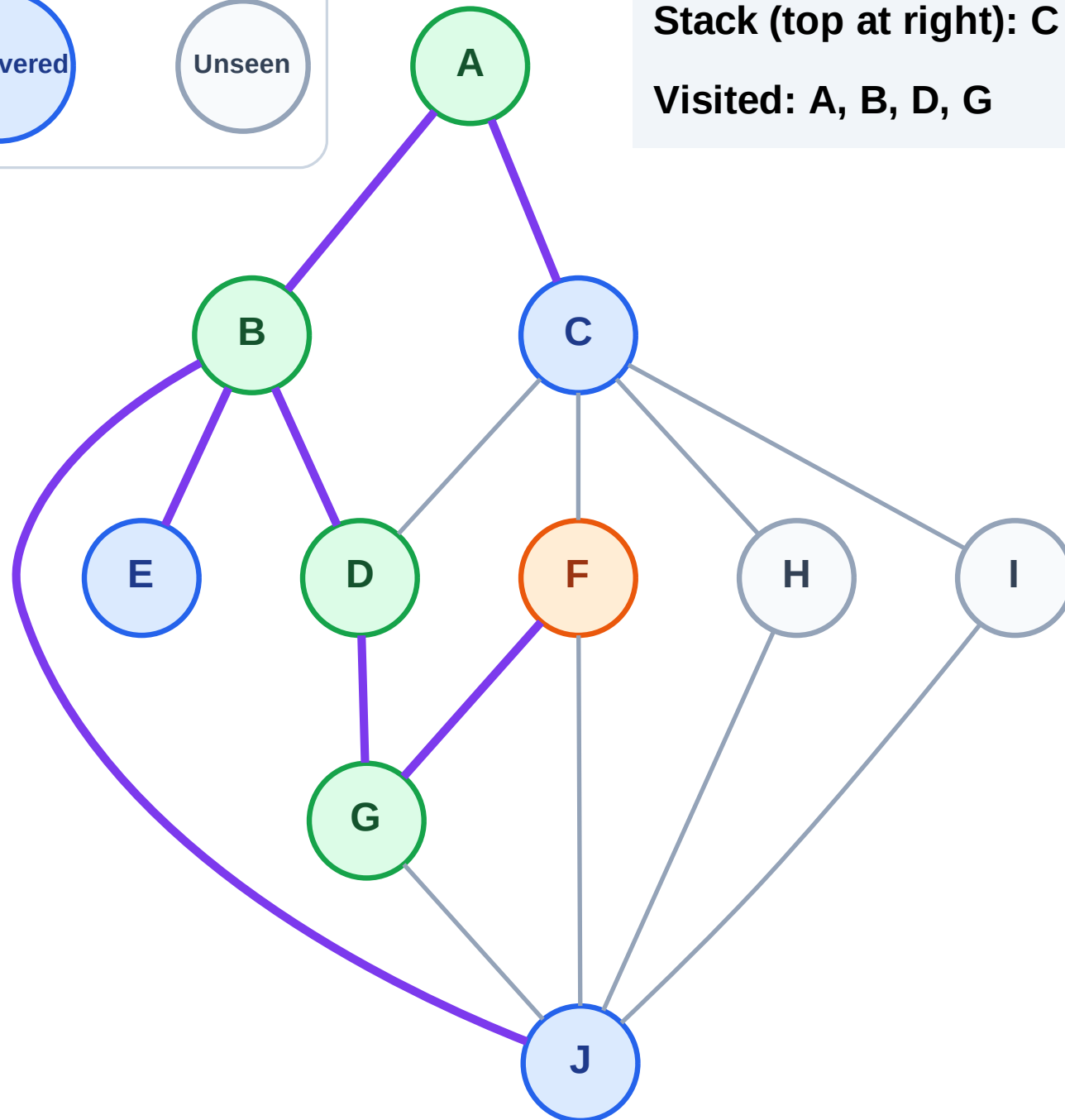
Step 10: Process node F

Legend



Stack (top at right): C | J | E

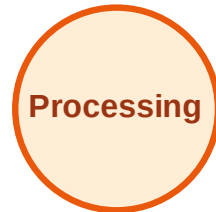
Visited: A, B, D, G



DFS Traversal

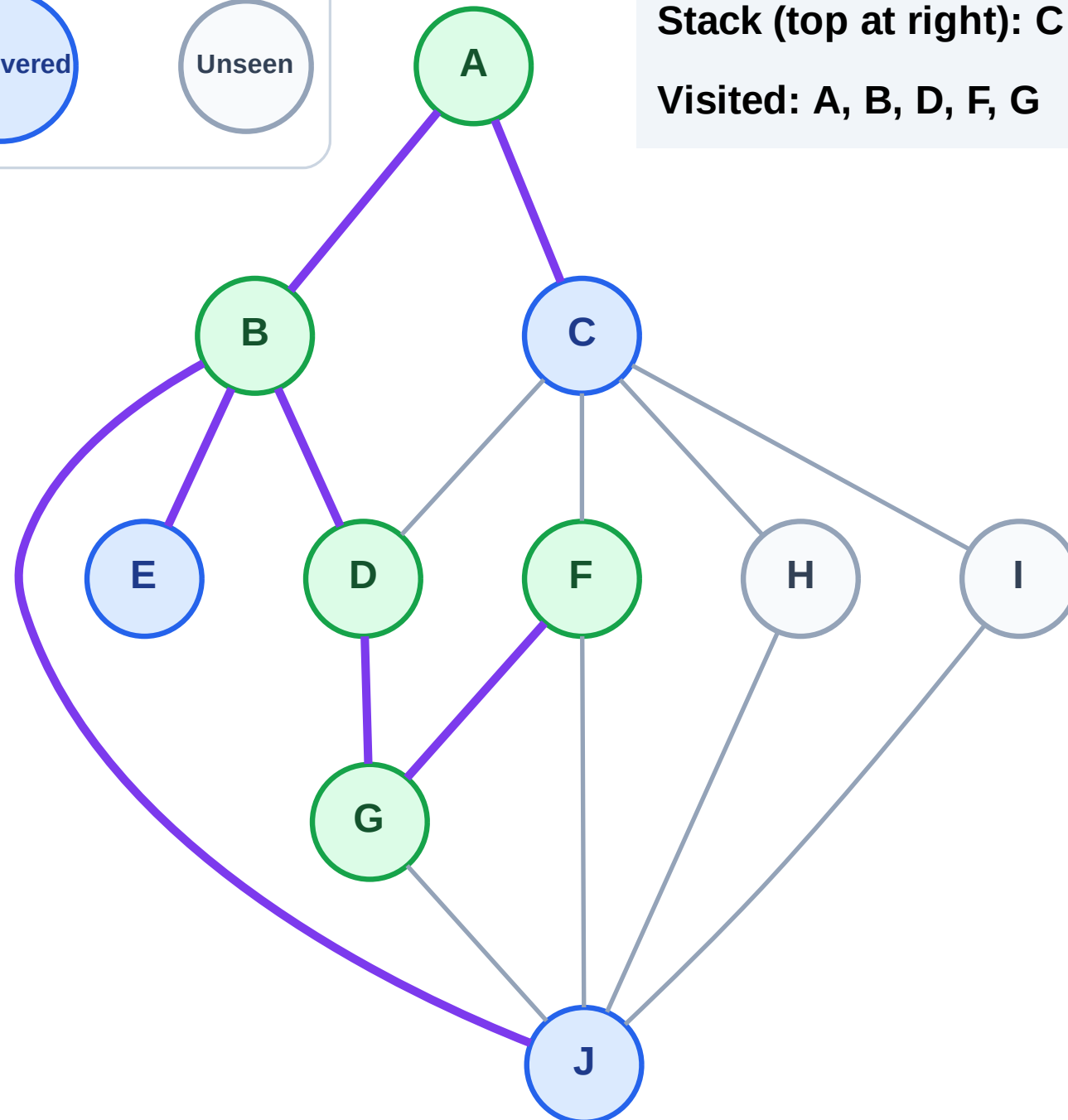
Step 11: No new neighbors from F

Legend



Stack (top at right): C | J | E

Visited: A, B, D, F, G



DFS Traversal

Step 12: Process node E

Legend

Complete

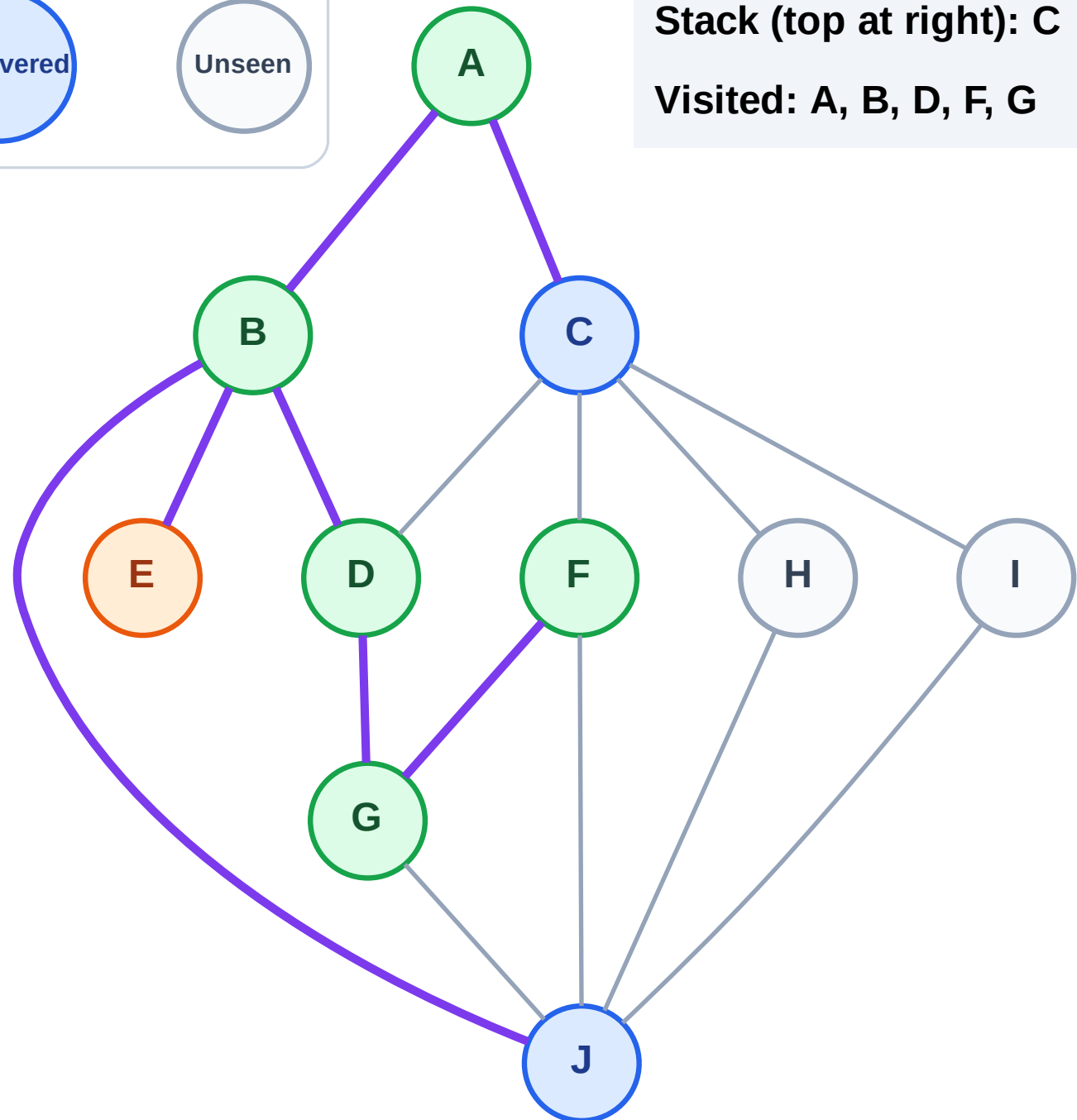
Processing

Discovered

Unseen

Stack (top at right): C | J

Visited: A, B, D, F, G



DFS Traversal

Step 13: No new neighbors from E

Legend

Complete

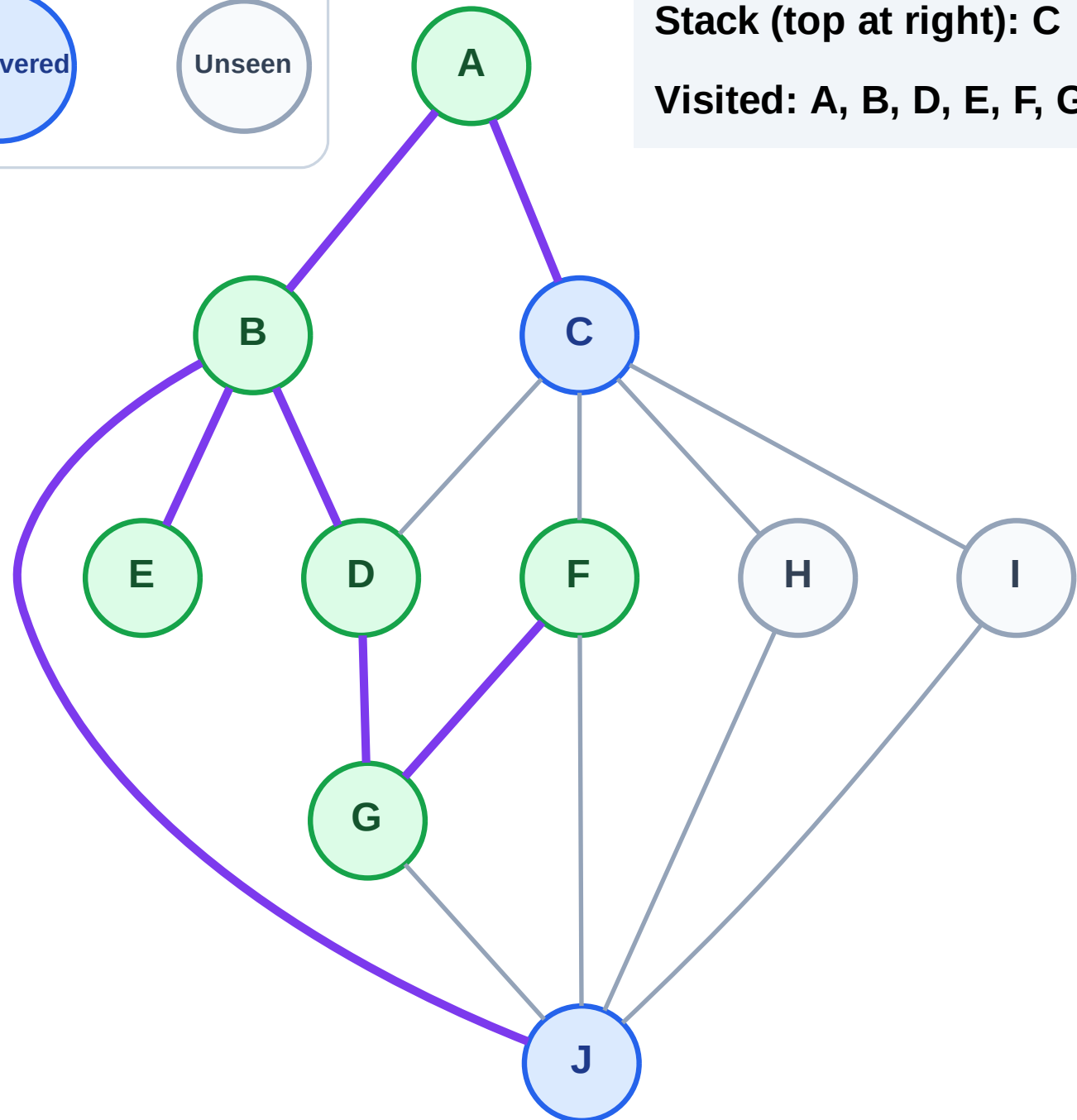
Processing

Discovered

Unseen

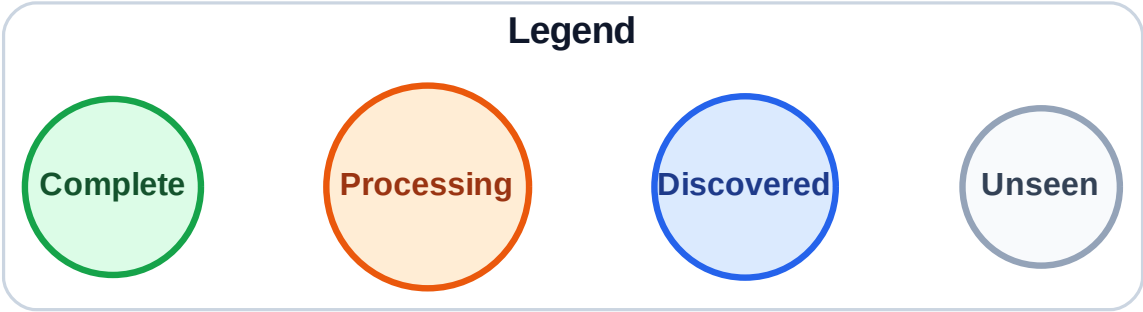
Stack (top at right): C | J

Visited: A, B, D, E, F, G

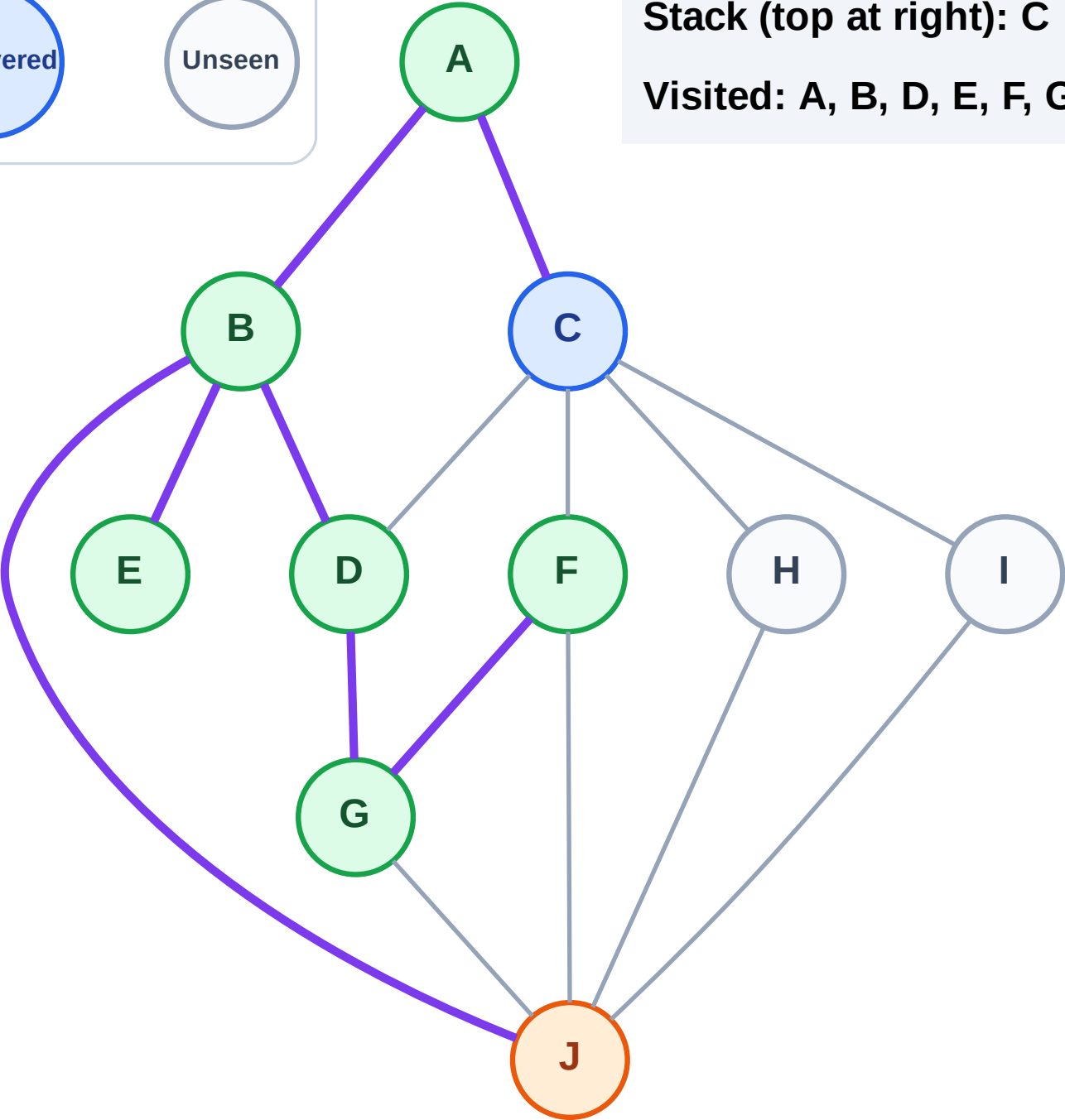


DFS Traversal

Step 14: Process node J



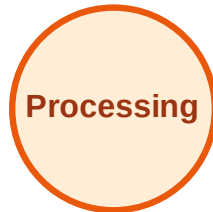
Stack (top at right): C
Visited: A, B, D, E, F, G



DFS Traversal

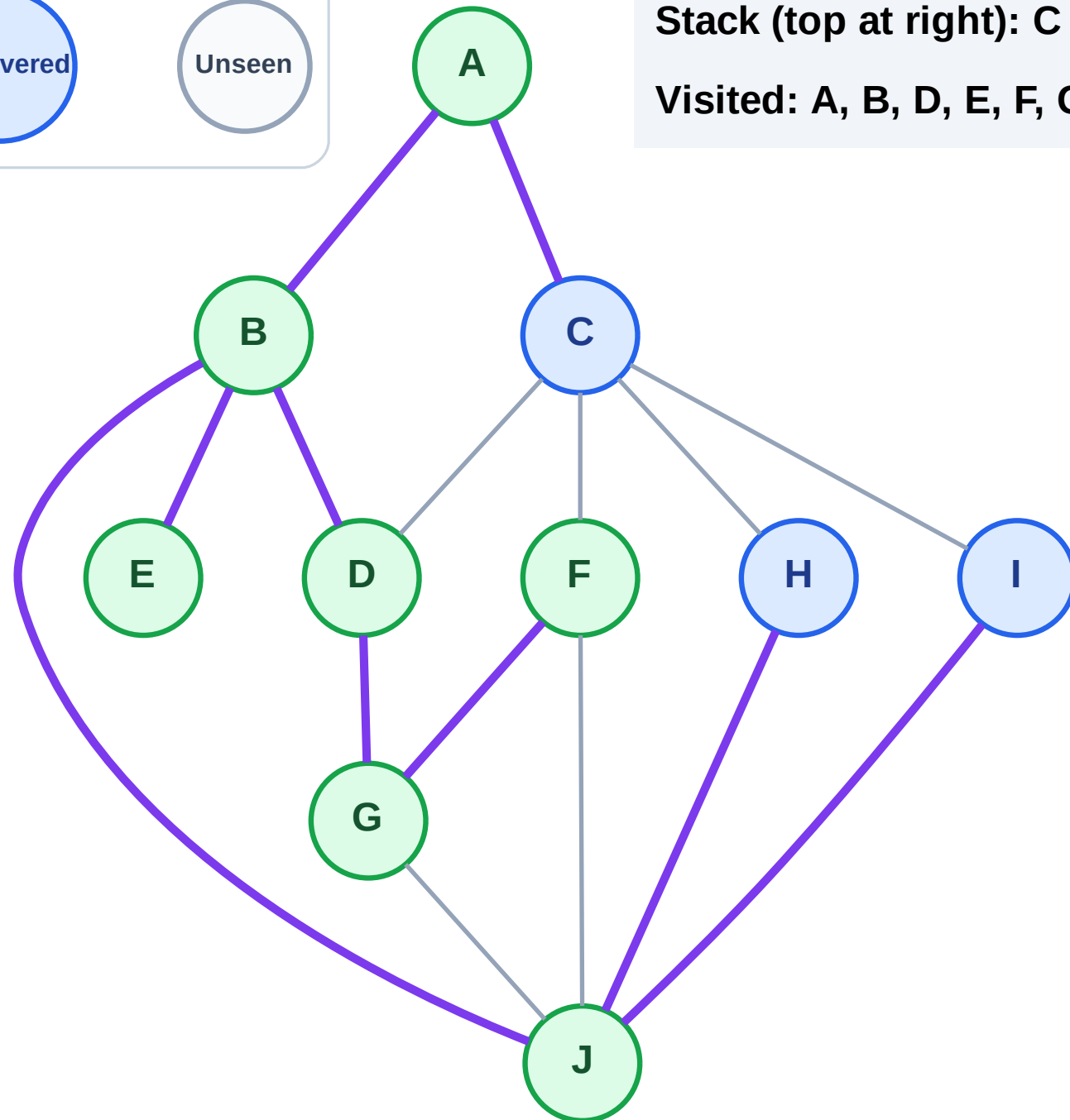
Step 15: Discover I, H from J

Legend



Stack (top at right): C | I | H

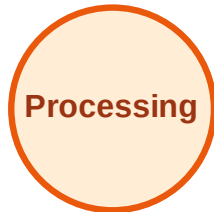
Visited: A, B, D, E, F, G, J



DFS Traversal

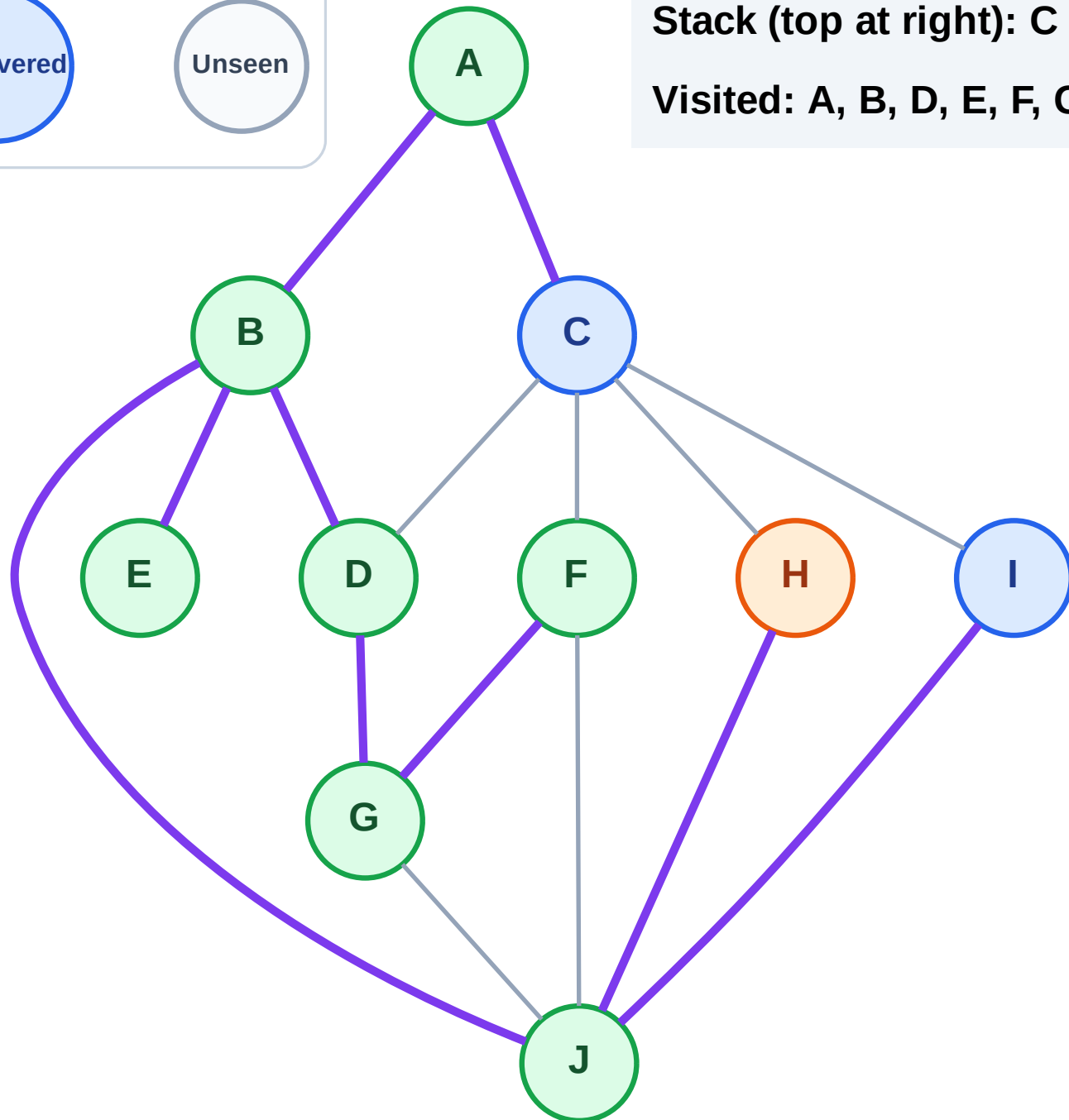
Step 16: Process node H

Legend



Stack (top at right): C | I

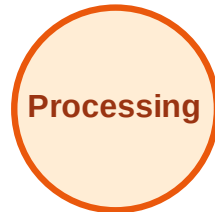
Visited: A, B, D, E, F, G, J



DFS Traversal

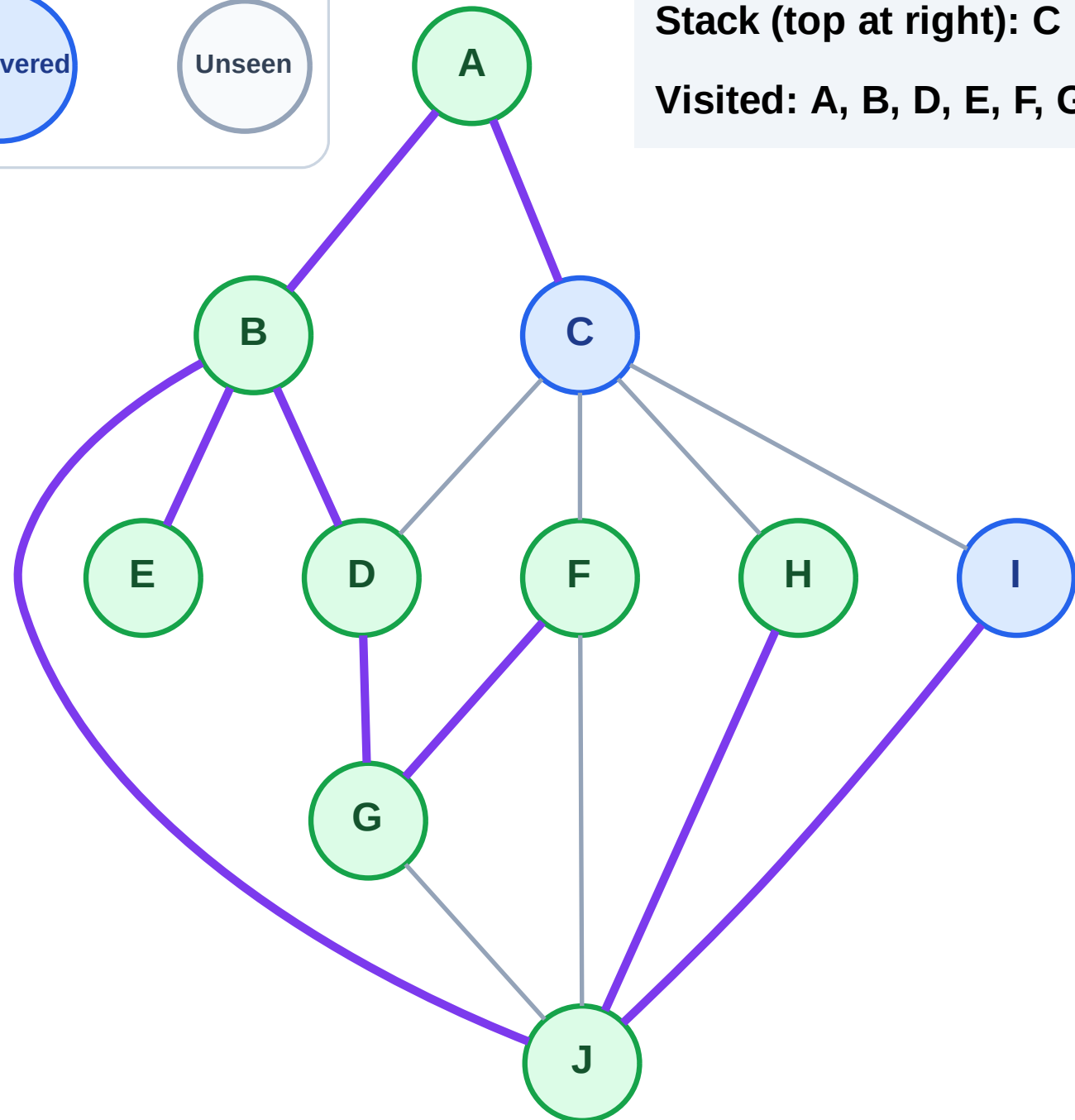
Step 17: No new neighbors from H

Legend



Stack (top at right): C | I

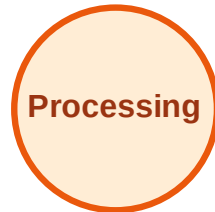
Visited: A, B, D, E, F, G, H, J



DFS Traversal

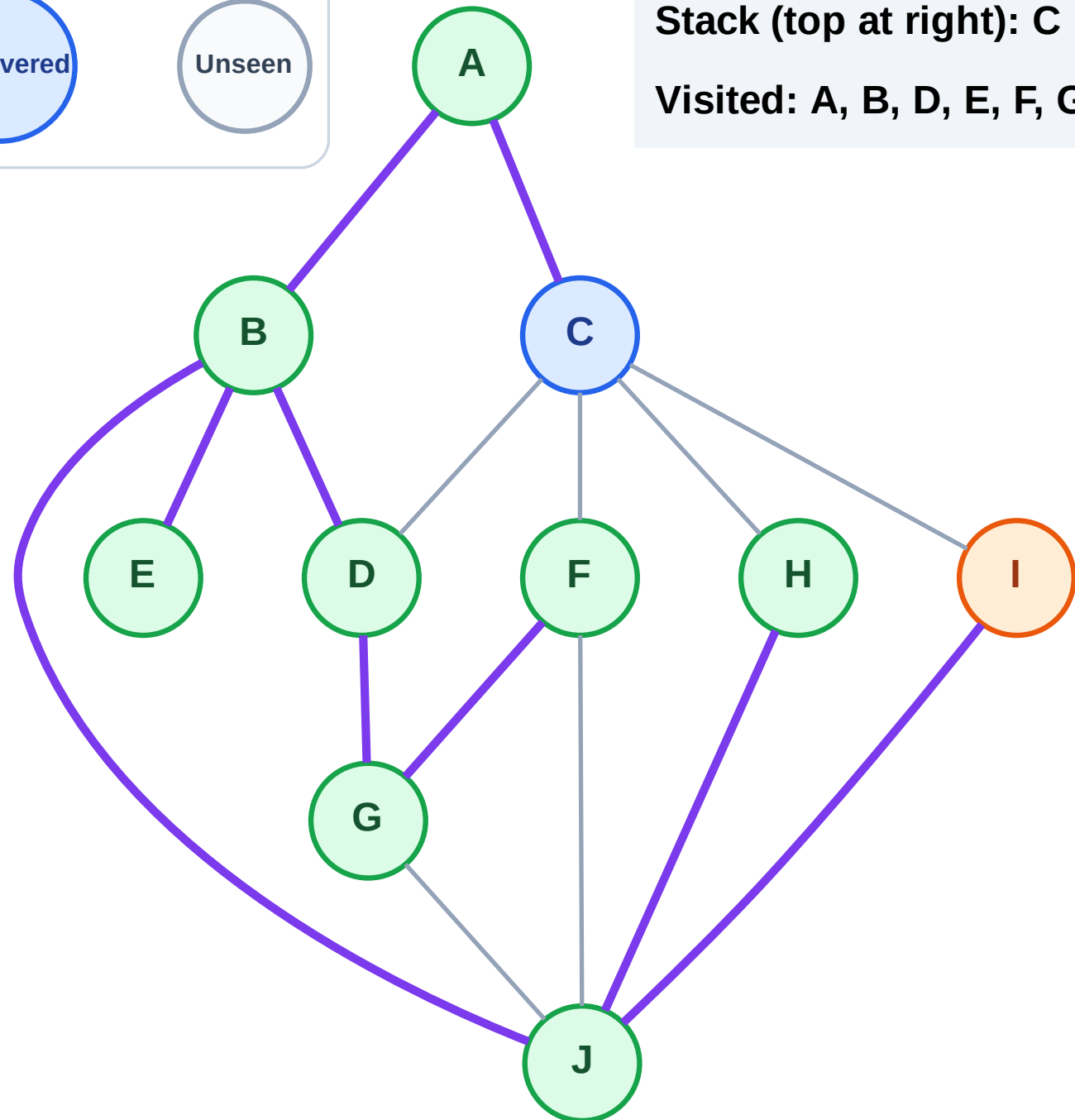
Step 18: Process node I

Legend



Stack (top at right): C

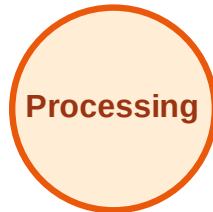
Visited: A, B, D, E, F, G, H, J



DFS Traversal

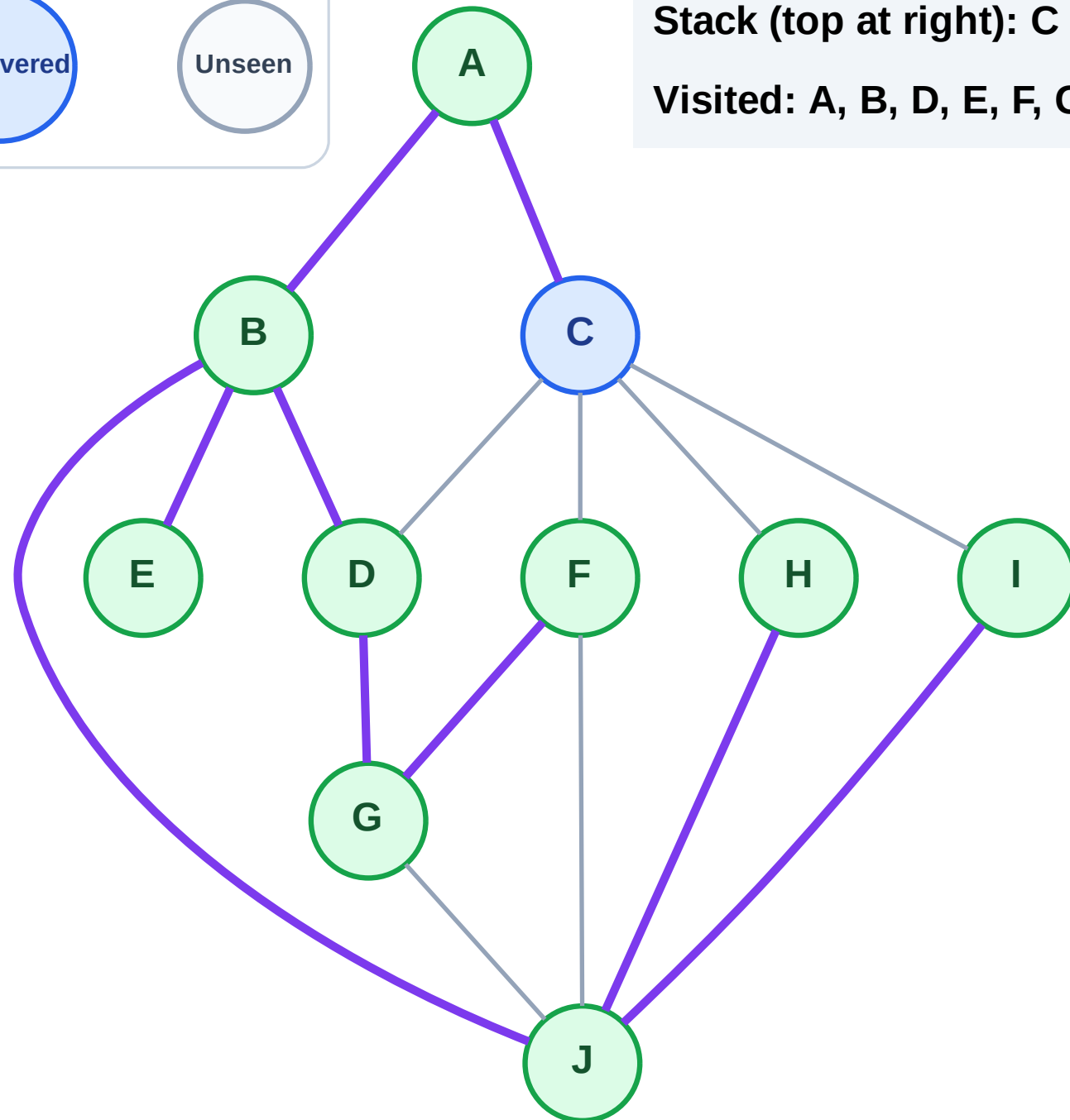
Step 19: No new neighbors from I

Legend



Stack (top at right): C

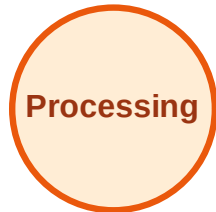
Visited: A, B, D, E, F, G, H, I, J



DFS Traversal

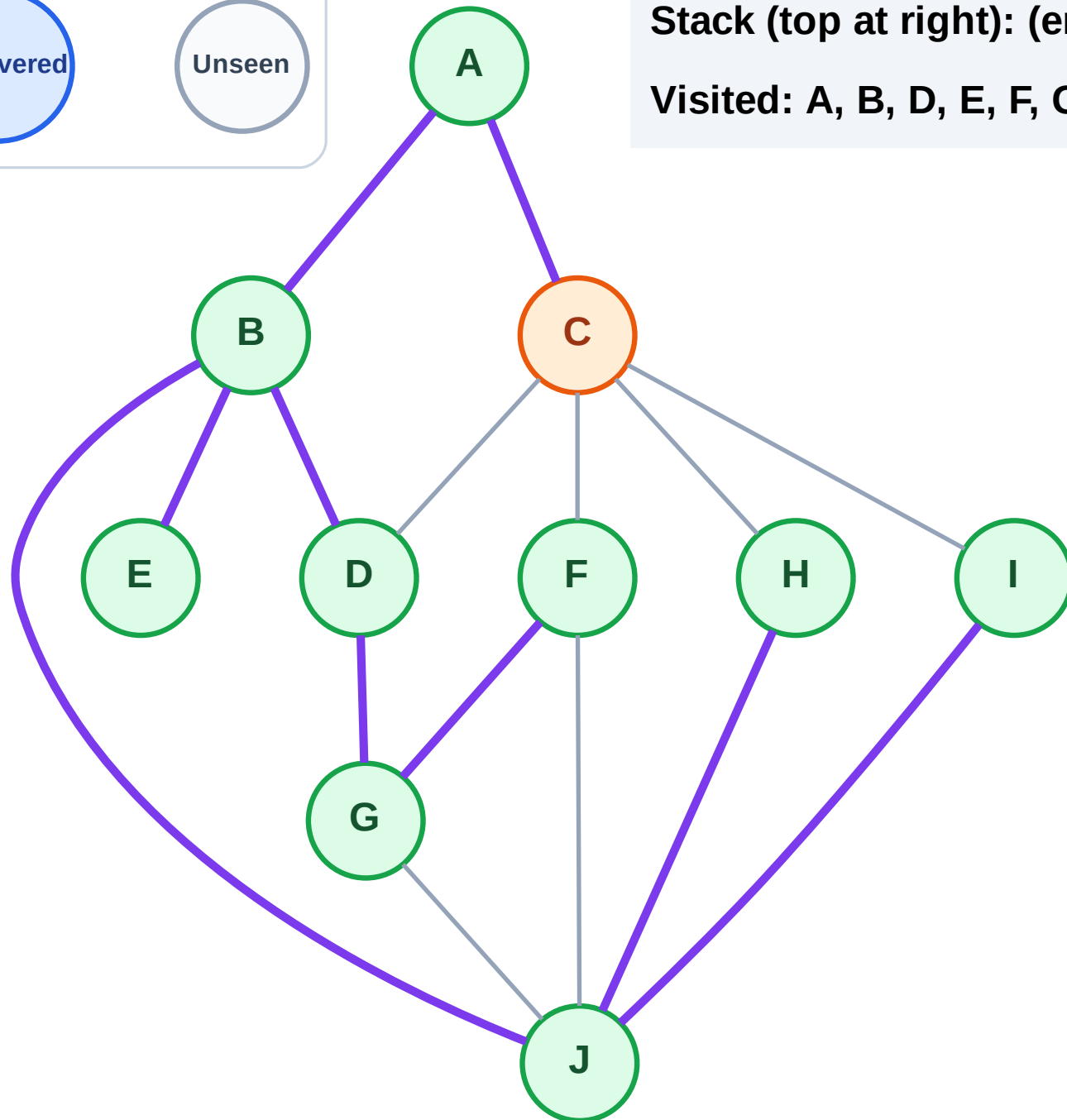
Step 20: Process node C

Legend



Stack (top at right): (empty)

Visited: A, B, D, E, F, G, H, I, J



Step 21: No new neighbors from C

Step 21: No new neighbors from C

